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LA-ICP-MS Trace Element Composition of Sphalerite and Galena of the Proterozoic Carbonate-Hosted Morro Agudo Zn-Pb Sulfide District, Brazil: Insights into Ore Genesis

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Abstract: The metal-rich Vazante-Paracatu Mineral Belt, in central Brazil, hosts the Zn-Pb sulfide Morro Agudo District in the Mesoproterozoic (1.3–1.1 Ga) upper carbonate sequence of the Vazante Group. The Morro Agudo district is comprised of the Morro Agudo deposit and the Bento Carmelo, Sucuri, and Morro do Capão occurrences. This carbonate sequence also hosts the Fagundes, Ambrósia and Bonsucesso Zn-Pb sulfide deposits (northern part) and the zinc silicate Vazante and North Extension deposits (southern part). The structurally controlled, stratabound and stratiform styles of mineralization in the Morro Agudo orebodies have been variably classified as sedimentary exhalative, Irish-type and Mississippi Valley-type. In this study, laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) spot analyses of sphalerite and galena from the Morro Agudo district revealed that red sphalerite (interpreted as the last stage) has higher Fe and Mn and lower Bi, Co, Cu, Ge, Hg, Tl compared to the other types of sphalerite, whereas the first generation of galena (Gn-I) is enriched in Ag, Cd, and Se and depleted in Cu and Mn relative to later galena (Gn-II). Mineral paragenesis and principal component analysis (PCA) of ore mineral composition suggest that the Morro Agudo, Sucuri and Morro do Capão mineralized zones formed by similar processes involving Zn-Pb mineralizing fluids with various compositions, comprising two main elemental associations: (1) Fe, As, In, Mn, Sb, Ag; and (2) Cd, Bi, Co, Ga, and Se. Bento Carmelo is distinguished in PCA by its unique dolomite-hosted sphalerite composition with elevated concentrations of Cu, Ge, Hg and likely formed from distinct fluids or processes. Temperatures of the mineralizing fluids for the Morro Agudo district range from 82 to 320 °C, calculated based on the trace element composition of sphalerite. The styles of mineralization and ore compositions are consistent with MVT deposits; however, fluid temperatures are hotter than typical MVT mineralizing fluids and may reflect a higher geothermal gradient or active advective fluid flow during the Brasiliano orogeny.

Keywords: carbonate-hosted sulfide Pb-Zn deposits; sphalerite and galena mineral chemistry; LA-ICP-MS; ore forming temperatures

1. Introduction

In Brazil, the bulk of base metal production is from the Vazante-Paracatu Mineral Belt in Minas Gerais, central Brazil. This mining district hosts Zn-Pb sulfide deposits (e.g., Morro Agudo, Fagundes, Bonsucesso, Ambrósia) and minor occurrences (e.g., Bento Carmelo, Sucuri, Morro do Capão) and world-class zinc silicate (e.g., Vazante, North Extension) deposits and occurrences (e.g., Pamplona, Cercado, and Olho d'Água: [1]). These deposits are hosted within carbonate rocks of the Proterozoic Vazante Group, which are interbedded with siliciclastic rocks.

The Vazante-Paracatu Mineral Belt has been the subject of numerous studies (e.g., [1–15]), and the proposed genetic models for the zinc silicate deposits, as well as the Fagundes and Ambrósia Zn-Pb sulfide deposits, are well constrained. Although various investigations

have been conducted on the Morro Agudo Pb-Zn deposit (e.g., [2–5,15–17]), its genesis is still controversial. The timing of mineralization is poorly constrained for the Morro Agudo deposit, and researchers have proposed that it is syn-sedimentary/syn-genetic (sedimentary exhalative (SEDEX) and Irish-types) or synchronous to the Brasiliano Orogeny (Mississippi Valley-type) (e.g., [5,18,19]) based on styles of mineralization (stratiform vs. stratabound), S isotope data in various sulfide minerals, and Pb isotope data in galena. Only a few studies have been conducted on the other sulfide mineral occurrences within the Morro Agudo district (Bento Carmelo, Sucuri, Morro do Capão) [15,20], and their relationship to the main orebodies of the Morro Agudo deposit is poorly understood.

Aldis et al. [15] have documented various generations of base metal sulfide and complex variations in sphalerite Fe and Cd contents within the Morro Agudo district, based on electron microprobe (EMP) analyses, and proposed that prolonged fluid mixing was involved in the formation of the various styles of mineralization, similar to other Mississippi Valley-type (MVT) deposits. However, most of the minor and trace elements were not detected by EMP.

Laser Ablation Inductively Coupled Plasma-Mass Spectrometry (LA-ICP-MS) allows for an analysis of most of the minor and trace elements at low concentrations that commonly substitute into the crystal lattice of sphalerite (Cd, Hg, Mn, Fe, Ga, Ge, In, Cu) and galena (Ag, Bi, Sb, Tl), and this information can provide insights into the fluid compositions and mechanisms and temperatures of precipitation (e.g., [21–38]). Furthermore, some elements (e.g., Ge, Ga, In: [39]) are considered essential for a low-carbon and digital economy [40], whereas others (As, Cd, Hg, Mn, and Tl in sphalerite; Cd, Sb, and Tl in galena) can be detrimental to the environment during processing [21,25]. Therefore, identifying their concentrations in the ore is relevant to ore processing and environmental assessment and protection.

This study reports the trace and minor element compositions of sphalerite and galena from the orebodies and occurrences in the Morro Agudo district. These results are compared to other zinc-lead sulfide deposits within the Vazante-Paracatu Mineral Belt and with the world-class zinc silicate Vazante deposit to better characterize the ore and understand the processes involved in their formation.

2. Geological Setting and Metallogeny of the Vazante Group

The Vazante-Paracatu Mineral Belt is hosted by the Vazante Group in the southern portion of the Brasília Fold Belt (BFB), which trends north–south for 1200 km parallel to the margin of the São Francisco Craton (SFC) [7,16,41]. The BFB was thrust eastward onto the SFC during the Brasiliano Orogeny (650–550 Ma) [41,42]. The ca. 800 km long SE-trending southern portion of the belt is composed of the Meso- to Neoproterozoic sedimentary Canastra, Ibiá, and Vazante groups that are thrust in a nappe system over the Bambuí Group [3,41,43].

The Vazante Group is a 250 km long arciform belt trending N–S (Figure 1) [3] that is composed of a dolomitic carbonate sequence with intercalated siliciclastic units. The stratigraphic sequence of the Vazante Group comprises seven formations from top to base (Figure 2) [3]: Lapa, Morro do Calcário, Serra do Poço Verde, Serra do Garrote, Lagamar, Rocinha, and Retiro-Santo Antônio do Bonito. The Retiro-Santo Antônio do Bonito and Rocinha formations (Lower Vazante Sequence) were interpreted as Neoproterozoic and the Lagamar to Lapa formations (Upper Vazante Sequence) as Mesoproterozoic based on detrital U-Pb zircon and organic-rich shale Re-Os dating [42,44,45]. Misi et al. [42] also identified a tectonic contact between the Lagamar and Rocinha formations where the Mesoproterozoic upper formations were interpreted to be thrust over the lower Neoproterozoic formations during the Brasiliano Orogeny.

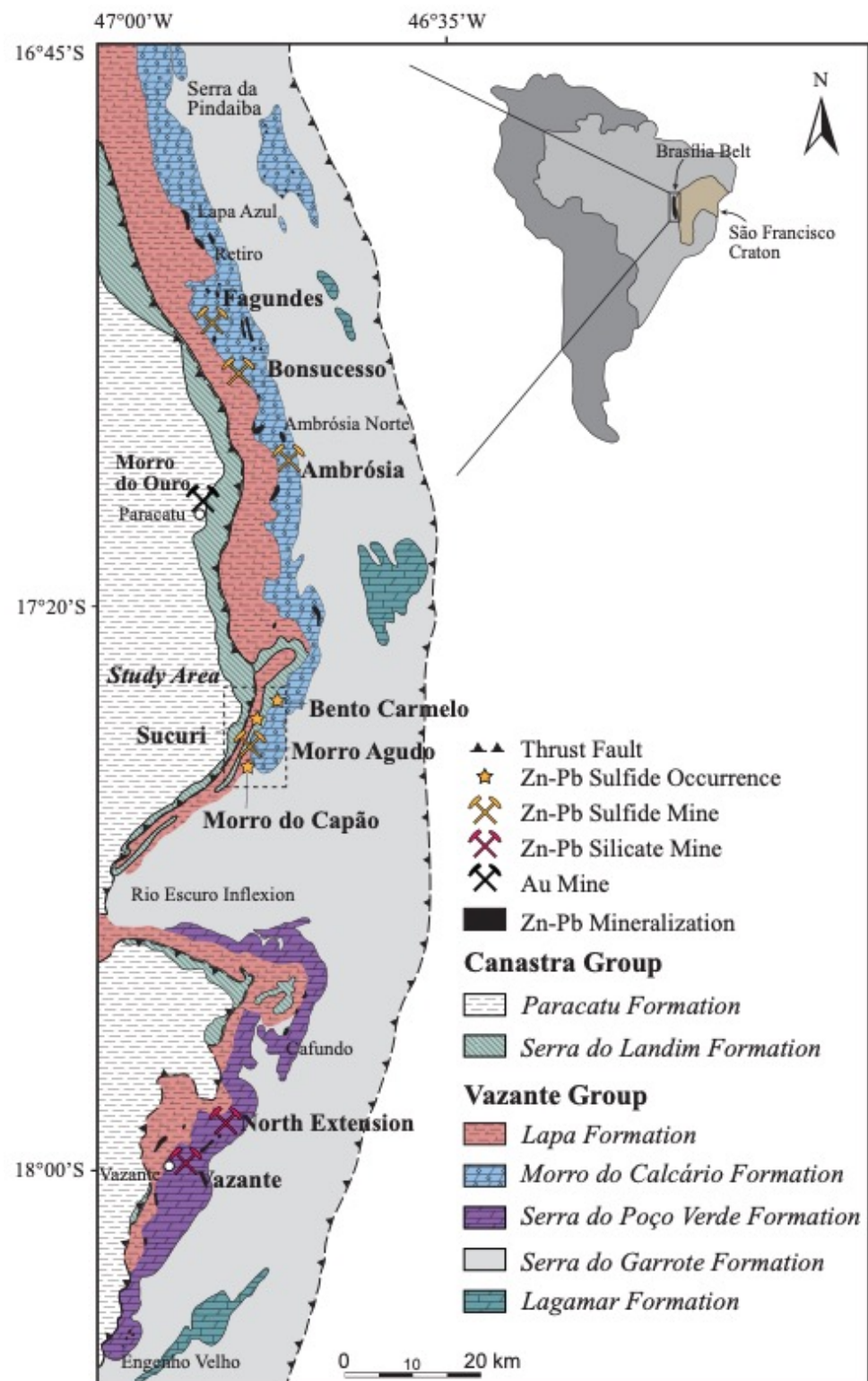


Figure 1. The geology of the Vazante-Paracatu mining district of the southern part of the Brasília Fold Belt, showing the locations of the mines and occurrences (modified from [6,11,12]), and the inset shows the relation to the Brasília Fold Belt with the São Francisco Craton. Dashed box represents the study area (Figure 3).

The depositional age and tectonic setting of the Vazante Group is ambiguous; researchers have proposed a passive margin setting, fore-arc orogenic domain, or a rift environment (e.g., [12,45–48]). Fernandes et al. [12] proposed that the source of detrital

material in the Vazante Group was Paleoproterozoic andesitic to felsic continental rocks, based on whole rock geochemistry, detrital zircon ages, and calculated Sm-Nd T_{DM} ages of 2.18–2.11 Ga [45]. The deposition age is interpreted to have occurred between 1.3 and 1.1 Ga, established by the youngest U-Pb detrital zircon concordant age of 1137 ± 8 Ma [45] and organic-rich shale Re-Os ages of 1100 ± 77 and 993 ± 46 Ma [44].

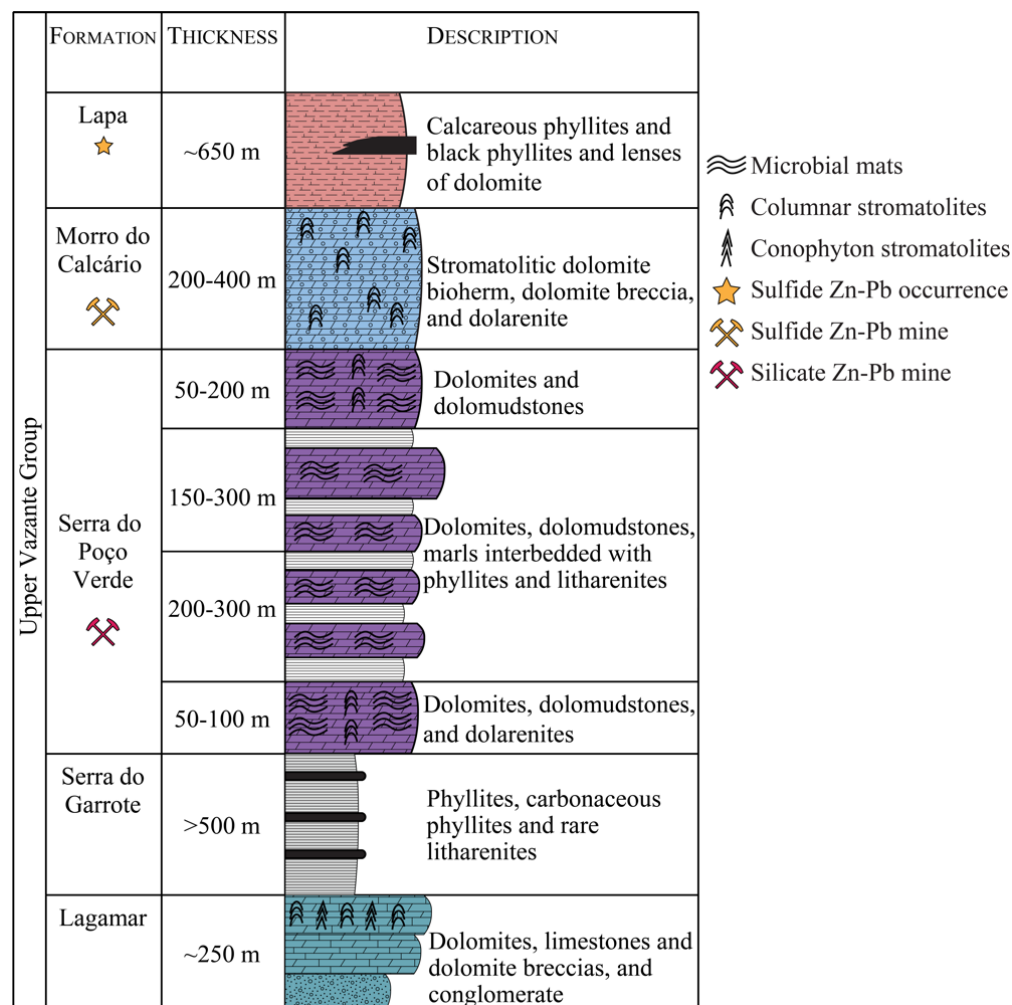


Figure 2. Upper Vazante Group stratigraphy showing the lithological controls on the various base metal deposits and occurrences (modified from [1,3,11,12,15]).

The Serra do Poço Verde dolomitic rocks, interbedded with thin siliciclastic rocks, are the main host of zinc silicate mineralization (e.g., Vazante and North Extension). However, the Morro do Calcário and Lapa formations host the Zn-Pb sulfide mineralization (e.g., Morro Agudo mine, Fagundes, Ambrósia, Bonsucesso and various minor occurrences investigated in this study). Oolitic, peloidal, and stromatolitic dolarenites and dolomites with varying degrees of brecciation are the predominant lithotypes in the Morro do Calcário Formation with thin layers of interbedded phyllite and siltstone units [3,7,15]. The Lapa Formation is composed of carbonaceous phyllites and metasiltstones with minor quartzite layers and lenses of stromatolitic and columnar dolomite [3]. These rocks are interpreted to have been deposited in an intertidal to subtidal environment [45]. The siliciclastic-rich Canastra Group overlies the Vazante Group and hosts the Morro do Ouro orogenic gold deposit [49] (Figures 1–3).

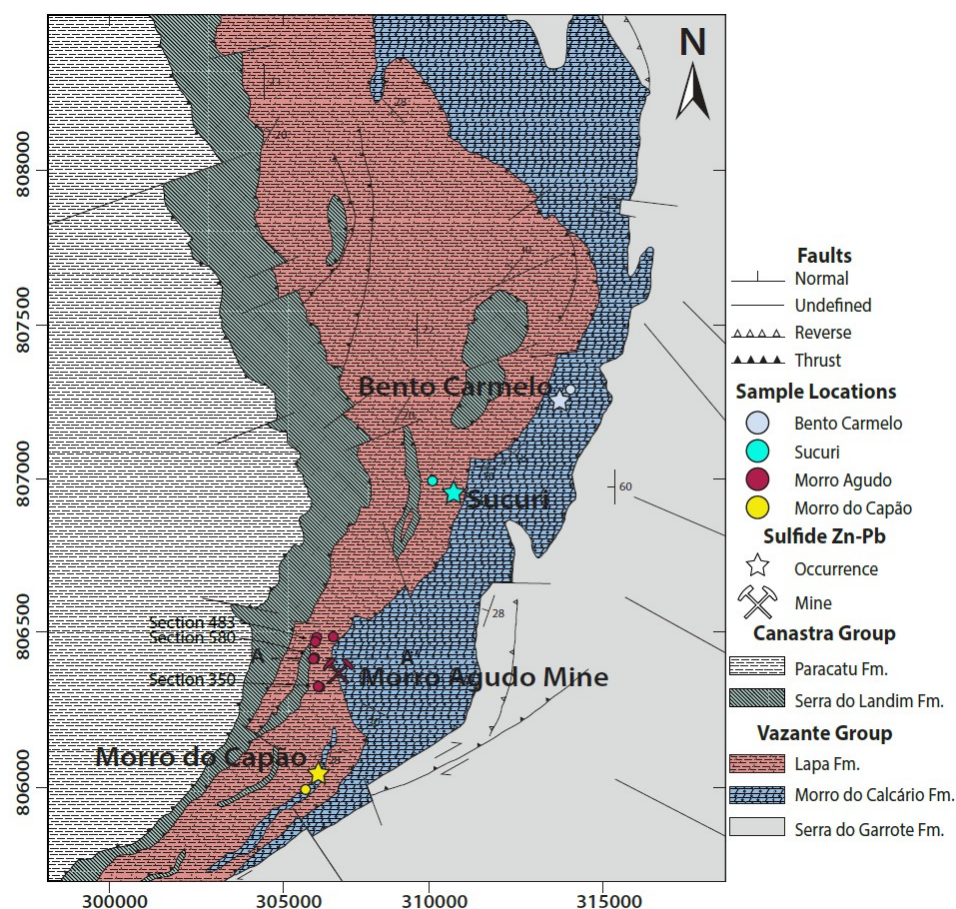


Figure 3. Local geology of the Morro Agudo district displaying the locations of the Zn-Pb deposit and occurrences and the locations of selected samples for this study (modified from [15,20] and internal Nexa database).

2.1. Morro Agudo District

The Morro Agudo deposit has reserves of 20 Mt and ore grades of 5% Zn and 2% Pb [11] and the resources of the proximal occurrences (Bento Carmelo, Sucuri, and Morro do Capão) have not yet been defined. Mineralization in the district is stratigraphically controlled in oolitic dolarenites, dolarenite breccias, and dolomite breccias, with subordinate mineralization in impure argillaceous dolomite and laminated dolomite [4,15,16,50]. Mineralization at Morro Agudo is also structurally controlled by a steeply dipping principal normal fault (N 15–20 W/75 SW) (Figure 4), with mineralization occurring in the hanging wall [5].

The Morro Agudo deposit is composed of multiple mineralized lenses, and the orebodies are denoted based on their stratigraphic position (from top to base: Figure 4): N, M, J, K, L, G, H, and I (GHI also referred to as Basal) [4,15,17,50]. J, K, and L are the main mineralized lenses and are grouped due to the similarities in host rocks, ore textures, and stratigraphic positioning [15]. G and H are composed of thin remobilized sulfide veinlets and are considered uneconomic for mining [4,15,16,50]. The structural geometry of the deposit is controlled by a normal fault system and zones (A, B, C, D, E) are delineated based on their position relative to the Principal Fault (Figure 4) [4,16]. Mineralization occurs in three sections, from north to south: 483, 580, and 350 (Figure 3).

Based on the variation of $\delta^{34}\text{S}$ values of sphalerite, galena, and pyrite from the M, JKL; GHI (Basal) orebodies (+10.6 to 38.8‰) and N orebody (−8.6 to +2.7‰), Misi et al. [16] proposed distinct genetic models (syngenetic for the N orebody and epigenetic for the others). Fluid inclusion studies of the orebodies at Morro Agudo documented varying mean salinities and homogenization temperatures throughout the deposit (4.9–13 wt.% eq. NaCl and 80–300°C; [4]), which Misi et al. [16] proposed to be similar to an Irish-type system.

Geochronological data suggest two ages of mineralization at Morro Agudo based on calculated Pb-isotope ages of galena: 1.1–1.0 Ga [19] and 650 ± 50 Ma [18]. Based upon these contrasting data, syn-sedimentary genetic models have been proposed for the N orebody [5,17], and epigenetic, syn-orogenic with the Brasiliano Orogeny for the main mineralized zones [5,16,17]; the latter is also supported by whole-rock Rb/Sr isochron ages of 680 ± 10 Ma of mixed pelitic-carbonate samples [51]. Based on textural and geochemical data, Aldis et al. [15] interpreted that mineralization was epigenetic and formed by similar processes, except within the Bento Carmelo occurrence, which has a unique composition that may be related to distinct processes or fluids.

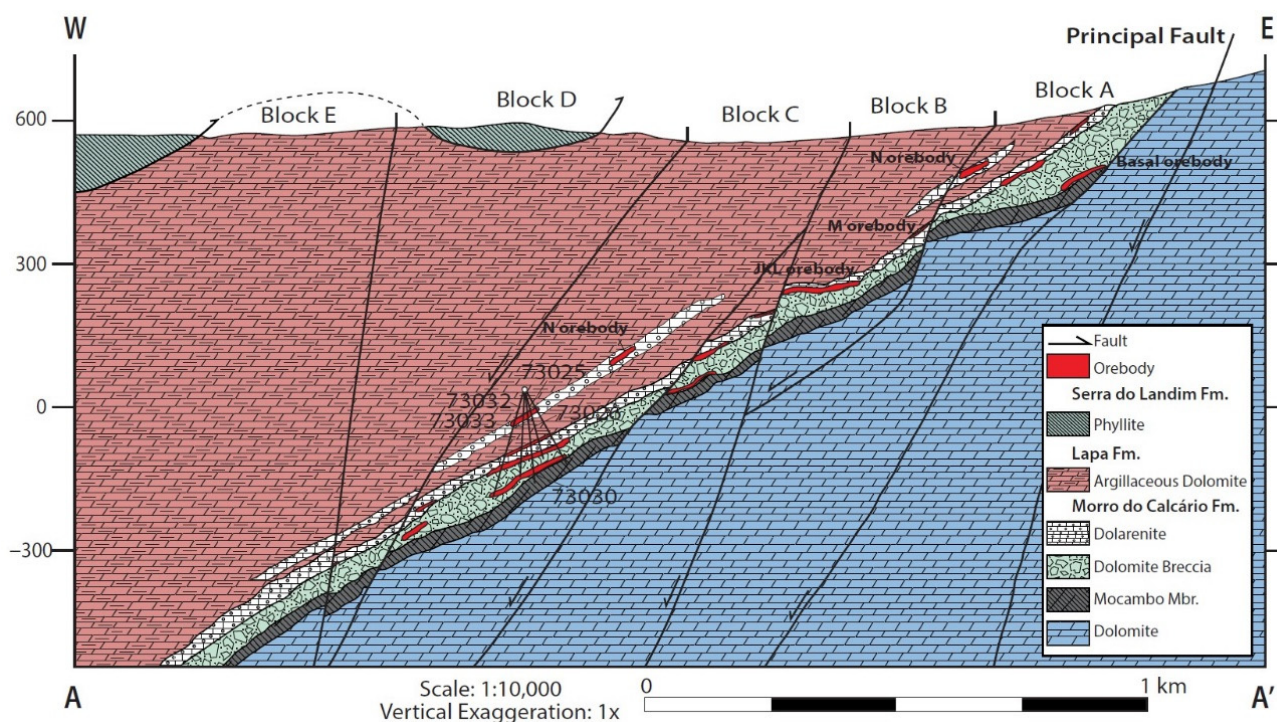


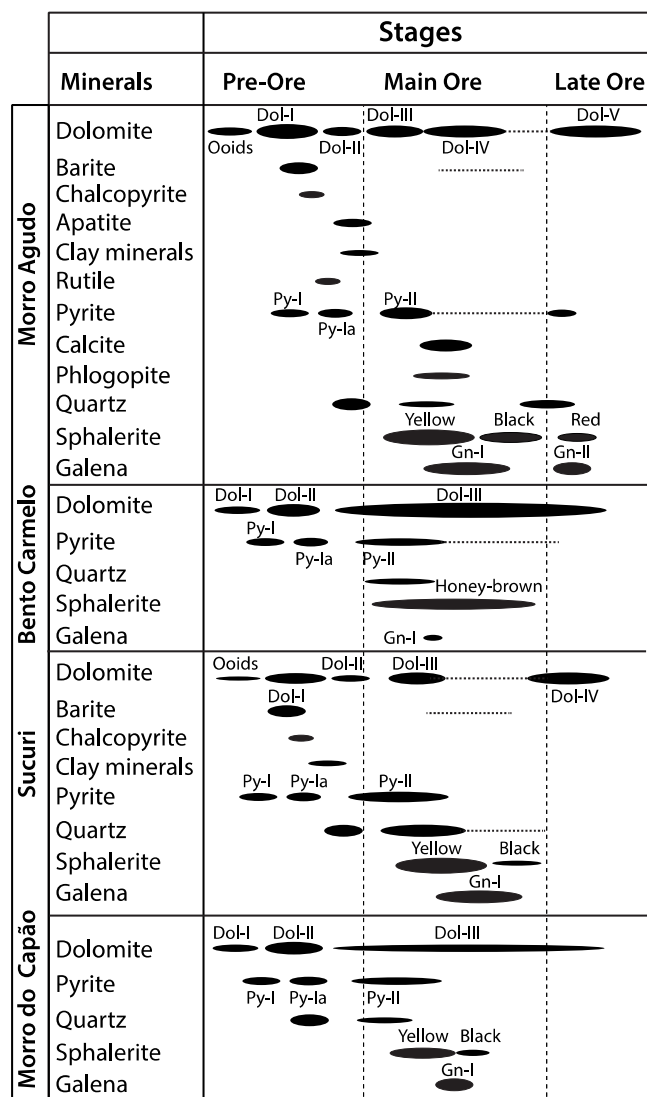
Figure 4. Cross-section of section 580 of the Morro Agudo deposit displaying the varying lithologies, stratigraphic positioning of the orebodies, location of drill holes, and orientation of structural features (modified from [4,15]).

Ore Mineral Textures, Compositions and Associated Hydrothermal Alteration of the Morro Agudo District

Various studies have described the mineralogy and textures in the distinct orebodies of the Morro Agudo deposit [15,17,50]. Aldis et al. [15] reported detailed mineralogical and textural relationships and chemical compositions of the various generations of sphalerite in the orebodies and minor occurrences in the Morro Agudo district. The relevant characteristics documented by Aldis et al. [15] are summarized in Table 1 and this section, and the interpreted paragenetic sequence is shown in Figure 5. The mineralization of the Morro Agudo district is characterized by various generations of sulfide minerals, mainly sphalerite, galena, and pyrite, with hydrothermal carbonate minerals and quartz filling corroded zones and veins in the host micritic and sparitic dolomite.

Table 1. Key characteristics of the host rocks and ore textures of the orebodies of Morro Agudo orebodies and other smaller Zn-Pb sulfide occurrences within the district.

Type	Basal Orebody	JKL Orebody	M Orebody	N Orebody	Bento Carmelo	Sucuri	Morro do Capão
Host Rocks	Dolarenite breccia, dolomite breccia	Dolarenite, dolarenite breccia, dolomite breccia	Dolarenite	Dolarenite	Dolomite	Dolarenite	Dolomite, argillaceous dolomite
Yellow–Yellow Brown	Breccia, disseminated, massive, cement, veinlets/vein	Massive, cement, brecciated, disseminated, veinlets	Sphalerite Textures—By Color Vein, veinlets, massive, disseminated	Laminated, disseminated, massive	Veins/veinlets, stringers, disseminated	Massive, cement, breccia, disseminated, veinlets	Veins/veinlets, stringers, disseminated
Black	Disseminated, cement, veinlets/veins	Disseminated, cement, veinlets/veins	Disseminated, veinlets	Disseminated	-	Disseminated, veinlets	Veins/veinlets, stringers, disseminated
Red	Veinlets/veins	Veinlets/veins	Veinlets/veins	-	-	-	-
1st	Veins/veinlets, disseminated, breccia	Veins/veinlets, massive, disseminated	Galena Textures—By Generation Vein, disseminated	Laminated, disseminated	Veinlets	Cement, disseminated, veinlets	Disseminated, open-space fill
2nd	Veins/veinlets, disseminated	Veins/veinlets, disseminated	Vein, disseminated	Disseminated	-	-	-

**Figure 5.** Simplified paragenetic sequence for the orebodies and occurrences within the Morro Agudo district (modified from Aldis et al. [15]).

Three main color types of sphalerite have been identified in the various Morro Agudo orebodies and other Zn-Pb sulfide occurrences in the district. Based on their relative timing, color, and textural relationships, three generations were identified (Figure 5) [15]. The earlier generations, sphalerite I (Sp-I) and II (Sp-II), are the most abundant and found throughout the district, and sphalerite III (Sp-III) occurs only in the Basal, JKL and M orebodies of the Morro Agudo mine.

The first generation of sphalerite (Sp-I) is coarse pale yellow to yellow-brown, texturally diverse, and is commonly associated with saddle dolomite, pyrite overgrowths, and quartz. In the M, JKL, and Basal orebodies of the Morro Agudo deposit and the Sucuri occurrence, Sp-I forms massive aggregates (Figure 6A), cements of allochems or clasts, veins/veinlets (Figure 6B), and infilling corroded pyrite veins or void-space (Figure 6C). Within the JKL and Basal orebodies, fine-grained, acicular phlogopite lathes are included in Sp-I [15]. In the upper N orebody, Sp-I occurs as laminations with pyrite, galena, and rare calcite overprinting chert and sparitic and micritic dolomite (Figure 6B).

At the Morro do Capão and Bento Carmelo occurrences, Sp-I commonly occurs within quartz-carbonate $\pm$ pyrite veins that cross-cut the host dolomite (Figure 6D). The pale sphalerite of Bento Carmelo commonly overprints pyrite-bearing and bituminous stylolites (Figure 6E) or is associated with disseminated pyrite and coarse saddle dolomite (Figure 6F).

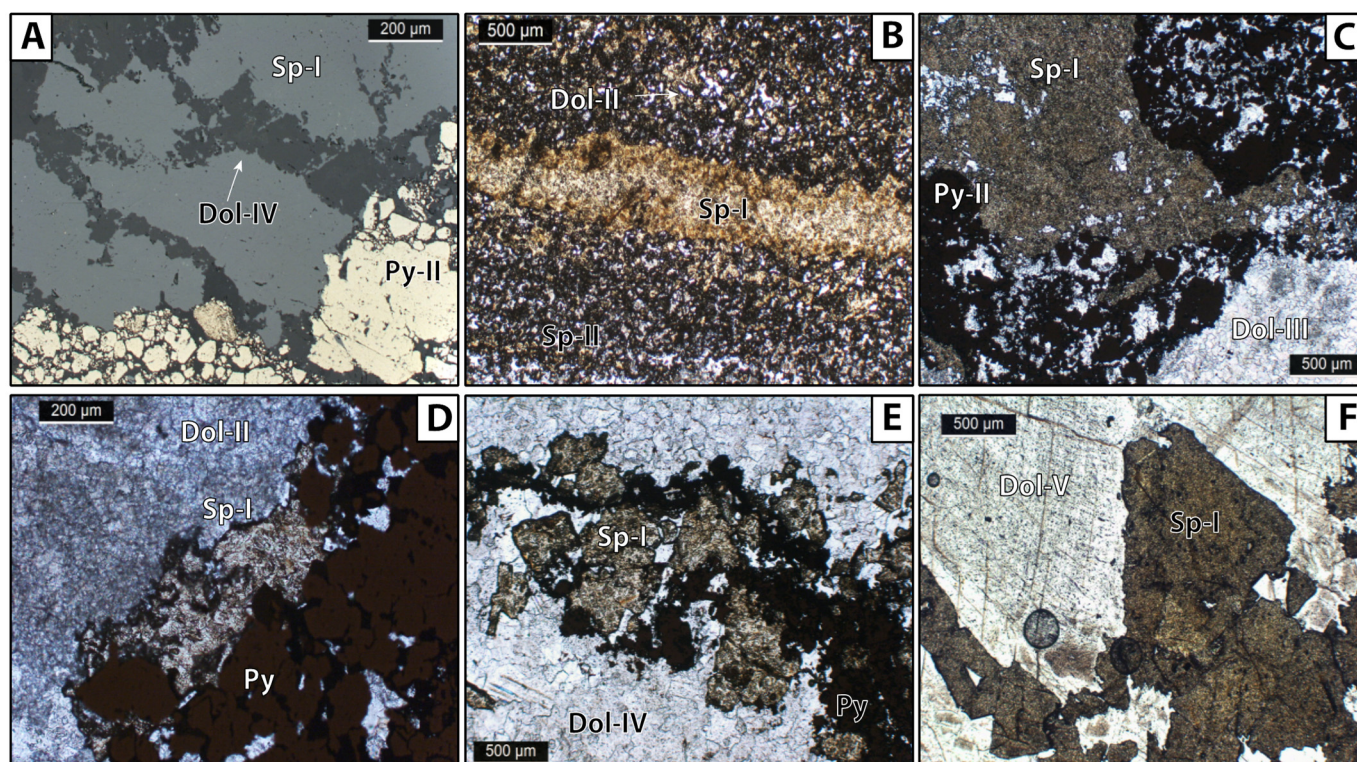


Figure 6. Examples of mineralized samples displaying textural and mineralogical features in the Morro Agudo district from the earliest stages of mineralization. Samples A and B are from the N orebody, C is from the Sucuri occurrence, D is from the Morro do Capão occurrence, and E,F are from the Bento Carmelo occurrence. (PPL = plane polarized light; RL = reflected light). (A) Coarse sphalerite (Sp-I) and dolomite (Dol-IV) associated with hydrothermal pyrite (Py-II) overgrowing earlier diagenetic pyrite (Sample ID = MA580-980-03; RL). (B) Sphalerite vein (Sp-I) and dolomite (Dol-II) being partially replaced by fine-grained disseminated sphalerite (Sp-II) (Sample ID= MA-N350-4; PPL). (C) Coarse vein of sphalerite (Sp-I) and pyrite (Py-II) cross-cutting hydrothermal dolomite (Dol-III) (Sample ID = S-PBC-45-31; PPL). (D) Yellow sphalerite filling space between coarse pyrite vein and sparitic dolomite (Dol-II) (Sample ID = MC-PMC-05-17; PPL). (E) Sphalerite replacing a pyrite-bearing stylolite (Sample ID = BC-MAN-34-04; PPL). (F) Coarse-grained sphalerite (Sp-I) vein associated with coarse hydrothermal dolomite (Dol-V) (Sample ID = BC-MAN-31-18; PPL).

The second sphalerite generation, Sp-II, is fine-grained and black and occurs throughout the Morro Agudo deposit and Sucuri occurrence (Figure 6B). It most commonly replaces the margins and rims of corroded or partially dissolved ooids (Figure 7A) or Sp-I. Additionally, Sp-II also occupies dissolution zones in sparitic dolomite or quartz and is cross-cut by Sp-III veinlets (Figure 7B). The last generation of sphalerite, Sp-III, is fine-grained, red,

and occurs only in the Morro Agudo orebodies, except in the N orebody. It occurs as late, cross-cutting veinlets (Figure 7B,C) and is commonly associated with late galena (Gn-II) (Figure 7D).

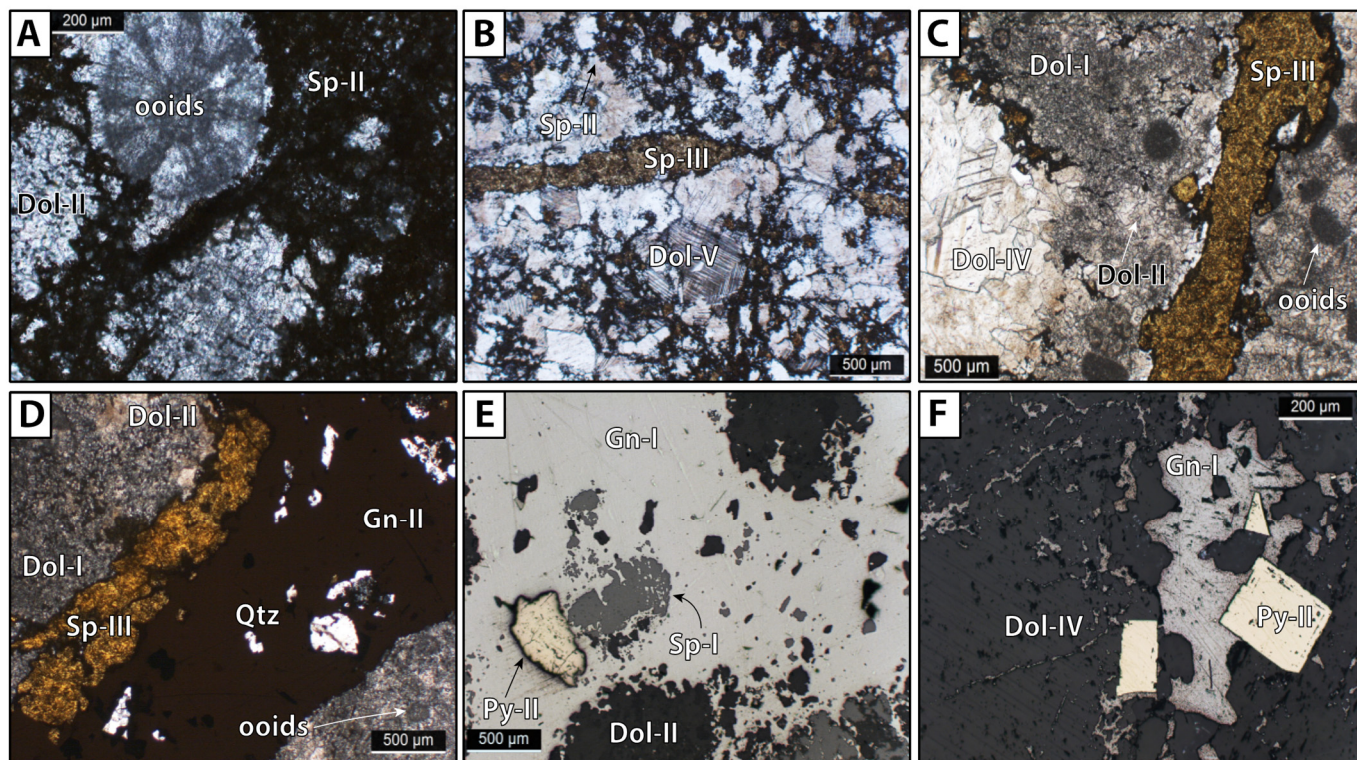


Figure 7. Photomicrographs of mineralogy and textural relationships of mineralized samples in the Morro Agudo district from the later stages of mineralization. Samples A,C,D,E are from the JKL orebody, B is from the Basal orebody, and F is from Morro do Capão (PPL = plane polarized light; RL = reflected light). (A) Fine-grained, black sphalerite (Sp-II) overprinting and replacing the margin of dissolved ooids and sparitic dolomite (Sample ID = MA580-033-30; PPL). (B) Veinlet of late sphalerite (Sp-III) cross-cutting coarse hydrothermal dolomite (Dol-V) and fine-grained black sphalerite (Sp-II) (Sample ID = MA580-030-28; PPL). (C) Veinlet of late sphalerite (Sp-III) cross-cutting sparitic (Dol-I) and micritic (Dol-II) dolomite with remnant ooids (Sample ID = MA580-033-30; PPL). (D) Late-stage galena (Gn-II) and sphalerite (Sp-III) cross-cutting micritic (Dol-I) and sparitic dolomite (Dol-II) and remnant ooids (Sample ID = MA580-053-30; PPL). (E) Coarse galena (Gn-I) filling corroded zones in sphalerite (Sp-I) and sparitic dolomite (Dol-II) (Sample ID = MA580-053-30; RL). (F) Interstitial galena (Gn-II) and euhedral pyrite (Py-II) infilling saddle dolomite (Dol-IV) (Sample ID = MC-PMC-05-20; RL).

Aldis et al. [15] identified five sphalerite compositional trends based on PCA analyses of the EMP results, including Zn, Fe, Cd and S. Trend 1 shows broad variability between Cd and Fe, encompassing the majority of the samples, including all from the M and N orebodies, most of the JKL and Basal orebodies, all of Sucuri, and one population from Bento Carmelo. Trends 2 and 3 are controlled by their Cd content: trend 2 has high Cd with samples from JKL (section 580), and trend 3 comprises samples from Bento Carmelo with low Cd (and detectable contents of Cu and Hg). Trend 4 is loaded by high Fe contents of samples from the JKL orebody, and trend 5 is composed of samples from the Basal (section 580) and JKL (section 350) orebodies and Morro do Capão occurrence with low abundances of Fe and Cd. Due to limitations of the EMPA, trace elements were below detection limit, and LA-ICP-MS allows the detection of the trace element mineral chemistry.

Two generations of galena are common throughout the orebodies of the Morro Agudo deposit, subordinate at Sucuri, and rare in the Morro do Capão and Bento Carmelo occur-

rences. The earlier galena phase (Gn-I) is more abundant, displays greater textural heterogeneity, commonly filling dissolution features in dolomite, pyrite-II, and Sp-I (Figure 7E), as skeletal or interstitial galena in corroded late carbonate (Figure 7F), or within veins, and is associated with coarse hydrothermal dolomite. The late galena, Gn-II, is found in cross-cutting veins/veinlets in association with Sp-III (Figure 7D). No trace element data were reported for galena from the Morro Agudo district.

3. Methodology

3.1. Sampling

Detailed logging was conducted on drill-cores from Nexa Resources' exploration program and mining operations that intersected sulfide mineralization in various lenses and located at distinct depths and distances in relation to the main fault at the Morro Agudo mine (Figure 3). Twenty-four drill-cores were selected across three sections (north to south: 483, 580, 350) of the Morro Agudo mine, four each from the Sucuri and Morro do Capão occurrences, and six from the Bento Carmelo occurrence. In order to determine the mineralogy and chemistry of the host rocks and mineralized zones, representative samples were collected for petrographic and whole-rock geochemistry analyses [15].

Eighty-seven samples were made into polished thin sections (PTS) from Morro Agudo, nine from Morro do Capão, nine from Sucuri, and fifteen from Bento Carmelo to examine the mineralogy, textural and paragenetic relationships between the altered and mineralized samples, and barren host rocks. A subset of twenty-two samples was selected for LA-ICP-MS trace element analysis to represent the different orebodies at distinct locations (depth and distance from the Principal Fault) in the Morro Agudo deposit and the occurrences. Samples from the greatest depth were collected from section 580, and drill-core from section 350 was the most proximal to the Principal Fault. Three samples each were collected from the N and M orebodies, five each from the JKL and Basal orebodies of the Morro Agudo deposit, and two from each of the minor occurrences in the district. The variation in abundance and generation of sulfide minerals, and the Zn-Fe-Cd sphalerite compositional trends identified by Aldis et al. [15] were considered in the selection of the subset of samples, in addition to grain dimensions due to the limitations of the laser spot size.

3.2. Laser-Ablation Inductively Coupled Plasma Mass-Spectrometry (LA-ICP-MS)

The trace element analyses of sphalerite and galena were conducted at the University of Ottawa's Laser Ablation Laboratory on a Photon Machines Analyte Excite 193 nm excimer ArF Laser Ablation system connected to an Agilent 7700x quadrupole ICP-MS. The laser system operated at a 11 Hz repetition rate using a beam spot size of 33 μm with a fluence of 3.54 J/cm². The ablation cell was flushed with 1 L/min He and then mixed with 0.75 L/min Ar. The analysis sequence consisted of 30 s background, 40 s analysis and then either 40 or 60 s between sphalerite and galena, respectively.

The following isotopes were measured: ²⁹Si, ³⁴S, ⁵⁵Mn, ⁵⁷Fe, ⁵⁹Co, ⁶⁰Ni, ⁶⁵Cu, ⁶⁷Zn, ⁶⁹Ga, ⁷⁴Ge, ⁷⁵As, ⁷⁷Se, ¹⁰⁹Ag, ¹¹¹Cd, ¹¹⁵In, ¹¹⁸Sn, ¹²¹Sb, ²⁰²Hg, ²⁰⁵Tl, ²⁰⁸Pb, and ²⁰⁹Bi. The dwell times for the respective elements are listed in Supplementary Materials (Table S1). A sequence of reference materials was inserted every 10 sample analyses; it consisted of USGS polymetal sulfide MASS-1 [52], synthetic glass standard GSE-1 G [53], and a pyrite.

Data reduction and drift correction for sphalerite were conducted using LADR v1.1.06 [54] using 100% normalization. The ⁵⁷Fe in reference material GSE-1 G was used for calibration of all elements except Zn, Hg, and S. Zinc was calibrated using MASS-1 as its Zn content is more comparable to that of sphalerite. Since S is not present in GSE-1 G, a pyrite was used to obtain the calibration for S using stoichiometric Fe and S values. Mercury has no certified concentration in GSE-1 G and thus MASS-1 was used instead. Silicon was measured to ensure that sulfide minerals were ablated.

Glitter v4.4 [55] was used for the galena data reduction using ²⁰⁸Pb as the internal standard on MASS-1 using stoichiometric concentration. The low abundance of Pb in MASS-1

relative to galena is a potential issue due to the matrix differences between sulfide minerals and the reference material, but analytical errors due to matrix-dependent fractionation is low, as discussed by George et al. [25]. Additionally, to verify that the interpolation on Pb was behaving according to expectation, the sum of all masses was checked against 100%. The internal calibration method does not a priori produce 100%, but since most expected galena components were measured, the sums were within 10% of 100% (mainly due to a correct estimation of S), and this was taken as a confirmation that the data reduction scheme was behaving correctly.

3.3. Data Treatment and Principal Component Analysis

Laser ablation profiles were inspected for any micro-inclusions (e.g., other sulfide or silicate minerals), and any samples with inclusions were excluded from data processing and analysis. Specific elements were excluded from the reported dataset as the majority (~>50% LoD) of samples were below the detection limit: Ni (72% < LoD) in sphalerite and As (65%), Co (84%), Cu (50%), Fe (70%), Ga (82%), Ge (71%), In (91%), Mn (51%), Ni (91%), and Zn (83%) in galena.

To assess the data properly, PCA was applied to the two LA-ICP-MS datasets (sphalerite and galena) in the Morro Agudo District to inspect the compositional relationships among trace elements. As PCA requires complete datasets, analyses below the detection limit were treated by random imputation with a normal distribution of the mean and standard deviation equal to the detection limit [56]. A centered log-ratio (clr) transformation was applied to the trace element composition prior to PCA to improve elemental associations [57] and reduce erroneous correlations [58]. Python was used for both the data treatment and the PCA.

4. Results

4.1. Sphalerite Trace Element Content

The compositional results of 570 sphalerite LA-ICP-MS spots are summarized in Table 2 (by orebody and occurrence) and Table 3 (by sphalerite color), and the full dataset is given in Supplementary Materials Table S2 (electronic supplementary material). The variation in composition in the various locations and generations is illustrated by boxplots (Figures 8 and 9), Cd-Fe binary diagram (Figure 10), and binary PCA plots (Figures 11 and 12). In addition to Zn and S, EMPA of sphalerite only consistently detected Cd and Fe, and the use of LA-ICP-MS allows for greater detection of trace elements that better constrains the ore conditions.

Iron and Cd are the most common and abundant trace elements in sphalerite, ranging from 0.14 to 3.54 wt.% and 0.15 to 0.63 wt.%, respectively (Figure 8), followed by Pb (district mean = 174 ppm), Hg (111 ppm), Mn (62.6 ppm), Cu (40.0 ppm), Sb (10.0 ppm), Ge (9.64 ppm), Ga (8.30 ppm), Ag (5.01 ppm), Se (1.39 ppm), Ni (1.16 ppm), Co (1.04 ppm), Sn (1.03 ppm), As (0.74 ppm), Tl (0.63 ppm), In (0.24 ppm), Bi (0.02 ppm) (Figure 8; Table 2). The orebodies of Morro Agudo and Sucuri occurrence showed similar concentrations of Cd, Mn, Pb, Sb, and Sn; however, As, Ag, Co, Cu, Ga, Ge, Hg, In, and Tl yielded more variable concentrations throughout the Morro Agudo district (Figure 8; Table 2).

The Cd content has the greatest variability in sphalerite throughout the orebodies of the Morro Agudo district compared to the minor occurrences, which have smaller ranges within the same occurrence. The highest mean concentrations of Fe (1.73 wt.%) and Cd (0.43 wt.%) are from the JKL and M orebodies, respectively, and the lowest concentrations in the district are from the Morro do Capão (Cd: 0.17%) and Bento Carmelo occurrences (Fe: 0.74 wt.%) (Figure 8A; Table 2).

Table 2. Trace element contents of sphalerite determined by LA-ICP-MS from the Morro Agudo deposit, and Sucuri, Bento Carmelo, and Morro do Capão occurrences. All units are in ppm unless otherwise stated.

Element	N (n = 108)				M (n = 63)				JKL (n = 88)				Basal (n = 165)			
	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD
Ag	0.02	0.92	0.20	95	0.15	40.2	11.4	100	0.06	30.7	4.88	99	0.07	51.3	5.79	100
As	0.16	1.92	0.67	81	<0.01	12.2	0.78	63	0.01	6.62	0.74	73	0.16	7.07	0.84	76
Bi	<0.01	0.07	0.01	65	<0.01	0.04	0.02	71	<0.01	0.04	0.01	70	<0.01	0.11	0.02	88
Cd (%)	0.26	0.58	0.41	100	0.31	0.61	0.43	100	0.21	0.62	0.41	100	0.15	0.63	0.38	100
Co	0.03	0.93	0.22	62	0.04	1.08	0.33	71	0.03	0.52	0.17	77	0.03	7.46	1.67	91
Cu	3.89	55.9	22.4	100	5.27	199	63.9	100	2.15	140	23.5	100	1.94	124	22.6	100
Fe (%)	0.75	2.92	1.52	100	0.18	2.37	0.96	100	0.20	3.54	1.73	100	0.64	2.25	1.31	100
Ga	0.25	10.1	1.50	100	0.25	194	20.0	100	0.43	60.7	6.11	100	0.52	47.0	6.17	100
Ge	0.07	2.84	0.63	62	0.12	167	37.6	75	0.09	33.3	2.56	81	0.06	42.7	1.93	83
Hg	11.8	139	46.0	100	39.6	112	69.0	100	5.58	107	33.1	100	4.26	108	34.3	100
In	0.01	0.03	0.02	6	0.03	3.27	0.56	90	0.02	1.06	0.18	84	0.01	0.56	0.09	70
Mn	11.1	935	92.0	100	11.5	313	38.9	100	6.40	483	59.4	100	3.20	949	51.4	100
Ni	0.39	0.53	0.48	6	0.49	7.72	1.56	17	0.46	7.21	1.27	11	0.31	9.65	1.04	52
Pb	10.7	1503	231	100	1.59	791	192	100	1.26	2193	212	100	0.66	1843	122	100
Sb	0.20	23.0	7.79	100	0.09	47.0	14.5	100	0.26	52.0	10.9	99	0.08	118	11.0	99
Se	0.47	1.77	0.92	62	0.88	6.43	2.28	87	0.52	1.79	0.94	58	0.44	5.46	1.55	87
Sn	0.46	1.78	0.86	77	0.67	3.20	1.26	56	0.51	2.16	0.94	77	0.46	8.39	0.99	94
Tl	<0.01	5.27	0.43	96	<0.01	5.87	0.67	90	<0.01	8.39	0.56	90	<0.01	4.85	0.19	93

Element	Sucuri (n = 50)				Bento Carmelo (n = 74)				Morro do Capão (n = 22)				Total Morro Agudo District			
	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD
Ag	0.77	51.0	10.3	100	0.17	5.85	2.47	100	0.05	1.57	0.46	100	0.02	51.3	5.01	99
As	0.19	2.48	0.56	58	0.13	0.85	0.33	36	0.22	0.96	0.42	68	0.13	12.2	0.74	68
Bi	<0.01	0.06	0.01	62	<0.01	0.05	0.02	78	<0.01	0.02	0.01	59	<0.01	0.11	0.02	74
Cd (%)	0.24	0.44	0.36	100	0.26	0.38	0.32	100	0.16	0.18	0.17	100	0.15	0.63	0.38	100
Co	0.03	2.63	1.06	72	0.08	8.42	1.76	100	0.07	0.08	0.08	9	0.03	8.42	1.04	78
Cu	4.60	117	26.9	100	5.80	297	115	100	9.00	163	33.1	100	1.94	297	40	100
Fe (%)	0.91	1.89	1.19	100	0.14	1.72	0.74	100	0.60	1.17	0.90	100	0.14	3.54	1.28	100
Ga	1.24	10.0	3.75	100	0.02	208	19.9	100	2.38	5.32	3.96	100	0.02	208	8.30	100
Ge	0.11	60.1	4.76	60	0.12	66.1	24.9	92	0.20	1.96	0.90	23	0.06	167	9.64	75
Hg	23.8	51.9	37.8	100	212	868	553	100	102	130	116	100	4.26	868	111	100
In	0.02	1.15	0.37	98	0.02	0.31	0.07	53	0.34	0.76	0.49	100	0.01	3.27	0.24	64
Mn	5.78	1290	82.1	100	1.45	444	67.8	100	5.96	59.9	19.7	100	1.45	1290	62.6	100
Ni	-	1.02	-	2	0.41	4.40	1.39	61	-	-	-	0	0.31	9.65	1.16	28
Pb	4.11	1371	194	100	0.25	718	144	100	7.59	1024	134	100	0.25	2193	174	100
Sb	0.13	33.8	7.62	100	0.04	28.4	4.29	85	3.77	49.2	18.6	100	0.04	118	10.0	98
Se	0.41	2.89	1.30	80	0.43	5.17	1.23	68	0.68	1.35	0.96	45	0.41	6.43	1.39	68
Sn	0.41	3.76	0.94	84	0.36	5.49	1.04	95	1.08	5.37	1.97	100	0.36	8.39	1.03	83
Tl	<0.01	6.20	0.74	90	<0.01	6.44	1.88	93	<0.01	4.62	0.54	95	<0.01	8.39	0.63	93

Table 3. Trace element contents of sphalerite by color. All units are in ppm unless otherwise stated.

Element	Pale Yellow (n = 67)				Yellow (n = 85)				Yellow-Brown (n = 323)			
	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD
Ag	0.24	24.2	3.64	100	0.02	22.6	1.98	96	0.02	51.3	5.81	99
As	0.13	0.98	0.39	40	0.18	6.26	0.87	81	0.15	12.2	0.66	68
Bi	<0.01	0.06	0.01	73	<0.01	0.08	0.02	65	<0.01	0.11	0.02	78
Cd (%)	0.26	0.59	0.35	100	0.17	0.60	0.41	100	0.15	0.63	0.36	100
Co	0.04	8.42	1.93	85	0.03	7.46	0.59	81	0.03	4.06	1.11	72
Cu	2.57	263	87.9	100	3.83	77.4	23.6	100	2.15	297	38.9	100
Fe (%)	0.14	2.24	0.83	100	0.87	2.92	1.51	100	0.18	2.37	1.21	100
Ga	0.23	155	13.7	100	0.26	25.7	2.42	100	0.02	208	9.90	100
Ge	0.08	66.1	19.6	84	0.08	6.17	0.89	71	0.06	167	11.9	73
Hg	5.58	868	380	100	6.44	139	34.1	100	4.26	654	99.1	100
In	0.02	0.56	0.13	61	0.01	0.37	0.07	28	0.01	3.27	0.31	73
Mn	1.45	444	60.8	100	9.86	814	74.5	100	5.78	1290	46.5	100
Ni	0.41	3.39	1.35	54	0.43	9.65	1.90	8	0.31	7.72	1.08	29
Pb	1.00	718	139	100	1.81	872	181	100	0.25	1673	152	100
Sb	0.04	45.1	5.48	84	0.34	33.1	9.97	100	0.08	118	9.71	99
Se	0.51	3.31	1.15	63	0.56	3.58	1.36	65	0.41	6.43	1.43	76
Sn	0.36	5.49	0.97	94	0.48	1.85	0.85	66	0.41	5.37	1.08	85
Tl	<0.01	6.44	1.41	93	0.01	6.20	0.39	88	<0.01	5.87	0.54	92

Element	Black (n = 52)				Red (n = 43)			
	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD
Ag	0.02	29.2	5.51	98	0.07	30.8	6.38	100
As	0.16	3.93	0.88	87	0.14	7.07	1.19	63
Bi	<0.01	0.11	0.02	81	<0.01	0.02	0.01	58
Cd (%)	0.17	0.54	0.40	100	0.30	0.61	0.44	100
Co	0.03	3.96	0.86	81	0.04	1.10	0.33	95
Cu	5.19	140	36.6	100	1.94	38.5	11.0	100
Fe (%)	0.64	2.14	1.40	100	0.99	3.54	1.84	100
Ga	0.44	22.1	4.12	100	0.52	12.6	4.46	100
Ge	0.10	33.3	2.87	79	0.09	4.07	0.60	77
Hg	5.92	100	31.5	100	16.57	70.3	29.4	100
In	0.02	1.00	0.12	42	0.02	0.63	0.13	95
Mn	3.20	949	129	100	11.37	483	81.6	100

Table 3. Cont.

Element	Pale Yellow (n = 67)				Yellow (n = 85)				Yellow-Brown (n = 323)			
	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD
Ni	0.39	3.68	0.92		33		0.42		2.24	1.30		12
Pb	7.73	2193	421		100		0.66		408	79.0		100
Sb	0.80	89.6	14.6		100		0.26		54.8	12.4		100
Se	0.52	4.11	1.44		73		0.52		3.91	1.44		79
Sn	0.46	8.39	1.16		90		0.54		1.63	0.91		79
Tl	0.01	8.39	1.01		98		<0.01		0.41	0.07		91

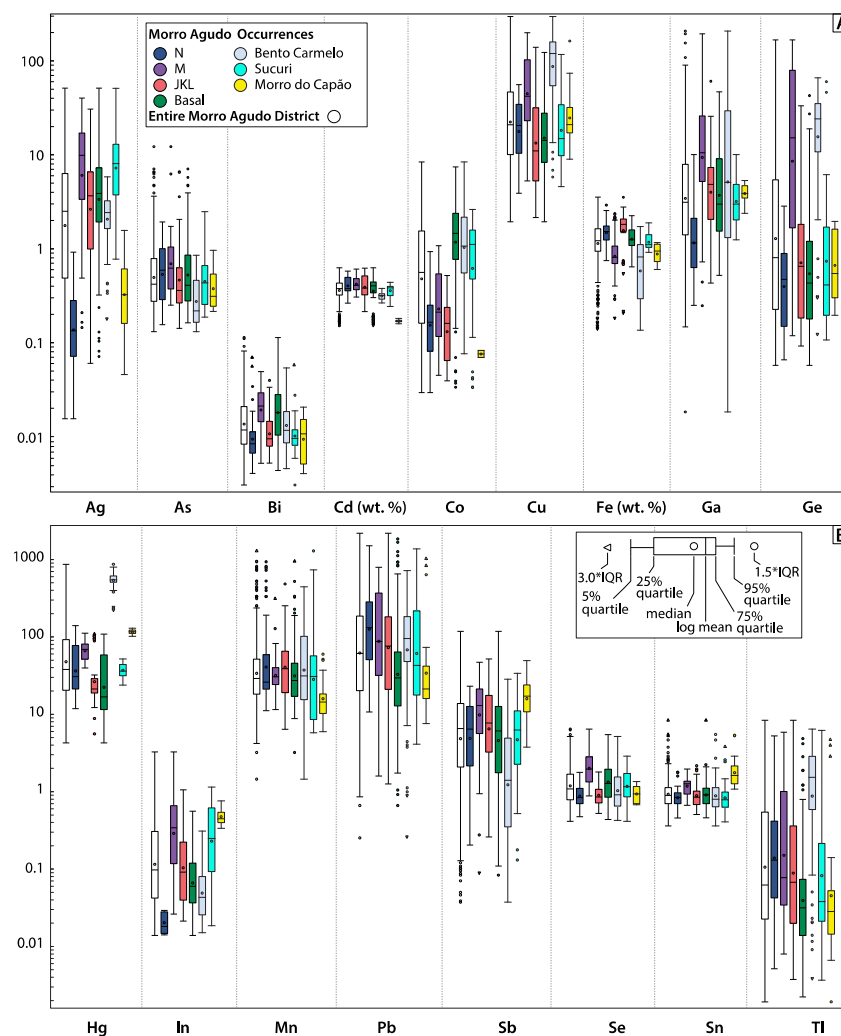


Figure 8. Box-and-whisker plots of trace elements (A) Ag, As, Bi, Cd, Co, Cu, Fe, Ga, Ge, (B) Hg, In, Mn, Pb, Sb, Se, Sn, and Tl in sphalerite within the Morro Agudo deposit and other district occurrences. All units are in ppm except Cd and Fe (wt.%). Note the logarithmic scale and corresponding mean value.

Bento Carmelo displayed enrichments of Co (mean 1.76 ppm; similar to the Basal orebody: mean = 1.67 ppm), Cu (115 ppm), Ge (24.9 ppm), Hg (553 ppm), and Tl (1.88 ppm) and lower contents in As (0.33 ppm), Fe (0.74 wt.%), and Sb (4.29 ppm) compared to others (Figure 8; Table 2). The Morro do Capão occurrence yielded elevated contents of Hg (116 ppm), In (0.49 ppm), Sb (18.6 ppm), and Sn (1.97 ppm) and lower concentrations of Ag (0.49 ppm, similar to the N orebody: mean = 0.20 ppm), Cd (0.17 wt.%), Co (0.46 ppm), Fe (0.90 wt.%, similar to the M orebody: mean = 0.96 wt.%), and Mn (19.7 ppm) compared with the rest of the district (Figure 8; Table 2).

Sphalerite was grouped by color (pale yellow, yellow, yellow-brown, black, red) and texture (disseminated, laminated, massive, open-space filling, replacement, stringer) to distinguish any variation between the interpreted generations (Sp-I, Sp-II, Sp-III). Typically, red sphalerite (interpreted as Sp-III) is commonly more enriched in Fe and Mn and relatively poorer in other trace elements (Bi, Co, Cu, Ge, Hg, Tl) compared to the other colors (Figure 9; Table 3). The sphalerite colors are generally compositionally similar (Figure 9) and show broad variability in their Fe-Cd ratios (Figure 10B–F).

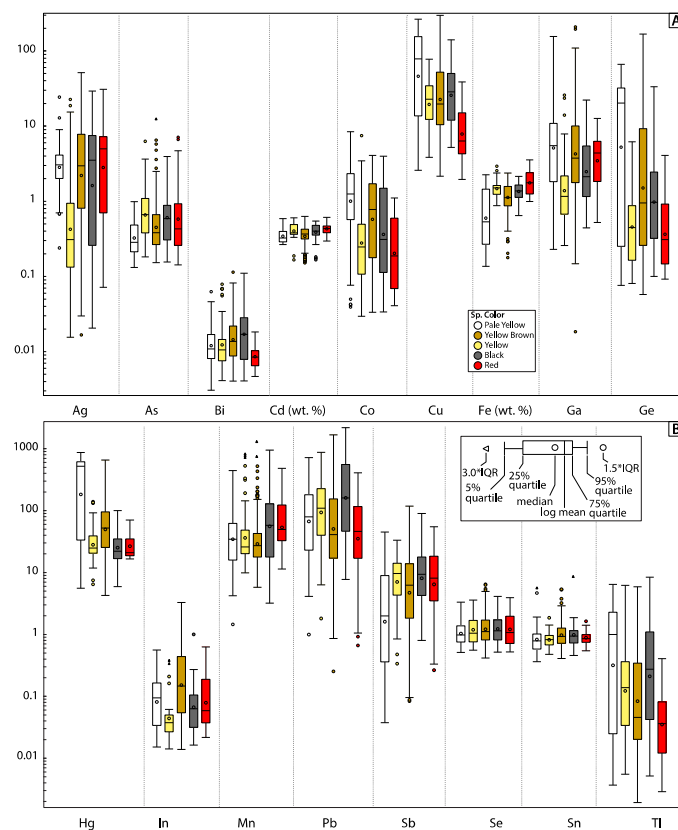


Figure 9. Box-and-whisker plots of trace elements (A) Ag, As, Bi, Cd, Co, Cu, Fe, Ga, Ge, (B) Hg, In, Mn, Pb, Sb, Se, Sn, and Tl in sphalerite distinguished by color. All units are in ppm except Cd and Fe (wt.%). Note the logarithmic scale and corresponding mean value.

PCA of Sphalerite

Principal component analysis was used to identify elemental associations or spatial variability throughout the Morro Agudo district in two sphalerite compositional datasets. The first dataset includes all the elements (Figures 11 and 12), and a secondary subset includes the elements applied for geothermometry (Ag, Co, Cu, Cd, Fe, Ga, Ge, In, Mn; Figure 12), following the methodology of Frenzel et al. [27].

By examining the entire elemental database, the first two principal components (PC1 = 29.2%; PC2 = 13.0%) account for 42.2% of the variation in the dataset (Figures 11 and 12). Broad overlap exists among the various generations of sphalerite from the JKL and Basal orebodies, and Sucuri and Morro do Capão occurrences, ranging from PC1+, PC2− to PC1−, PC2+ through the origin, and loaded by Ag, As, Bi, Co, Cd, Fe, Ga, In, Mn, Pb, Sn, Se, Tl, and Zn. PC1− comprises the two most abundant trace elements that substitutes for Zn (i.e., Fe and Cd), but Fe is associated with Ag, As, In, Mn and Sb in PC2−, and Cd is associated with Bi, Co, Ga and Se in PC2+. Most of the Bento Carmelo sphalerite and little of the sphalerite from the M orebody (section 580) are located in PC1+, PC2+ and loaded by Cu-Ge-Hg (Figures 11 and 12). The distinct signature of Bento Carmelo sphalerite was also captured in PC3 and PC4 (Figures S1–S4—Supplementary Materials).

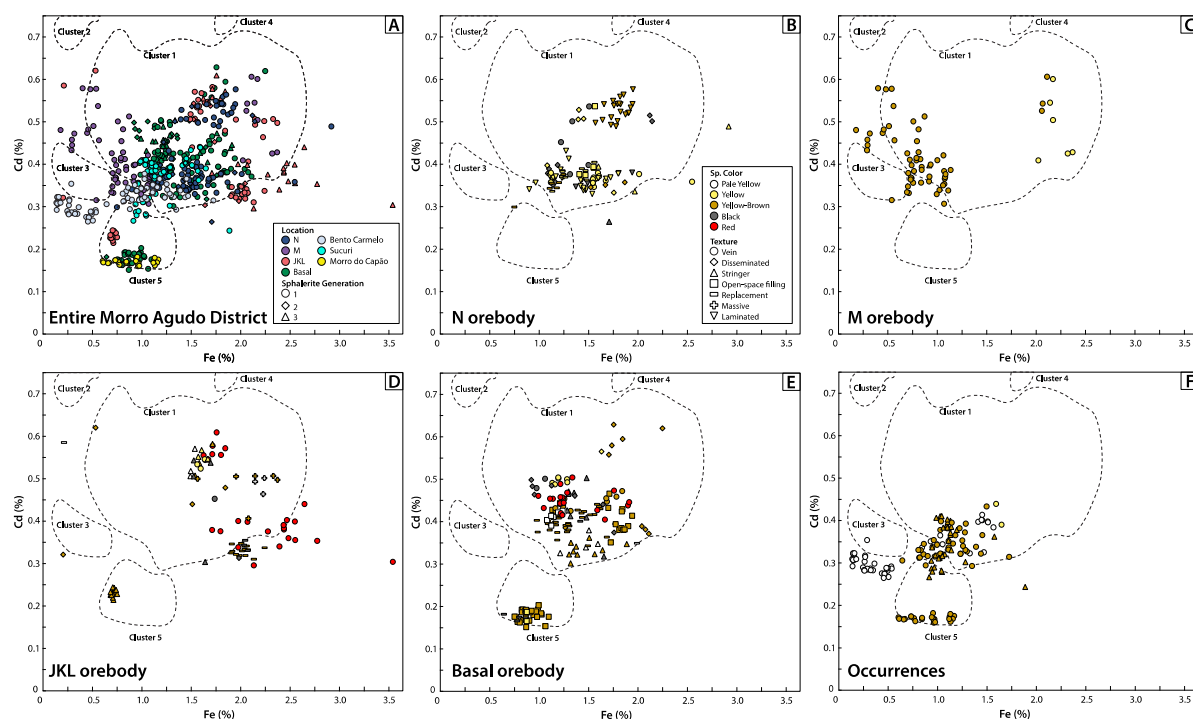


Figure 10. Binary diagram of Fe–Cd (%) concentrations determined by LA–ICP–MS in spherulite for the entire Morro Agudo deposit and occurrences (A) and individual orebodies and minor occurrences (B–F—legend in B) with compositional fields of EMPA spherulite data from Aldis et al. [15] shown for comparison.

PCA plots for individual orebodies and minor occurrences were constructed (Figure 12) and these reveal two general elemental associations: (1) Ag–Cu–Ge–Pb–Sb–Tl±As and (2) Cd–Co–Fe–Ga–Hg–Mn–Se–Sn±In. No observable patterns exist that are controlled by spherulite texture or color (Figure 12), spatial distribution in the different sections of Morro Agudo or the host rock type (Figure S5—Supplementary Materials).

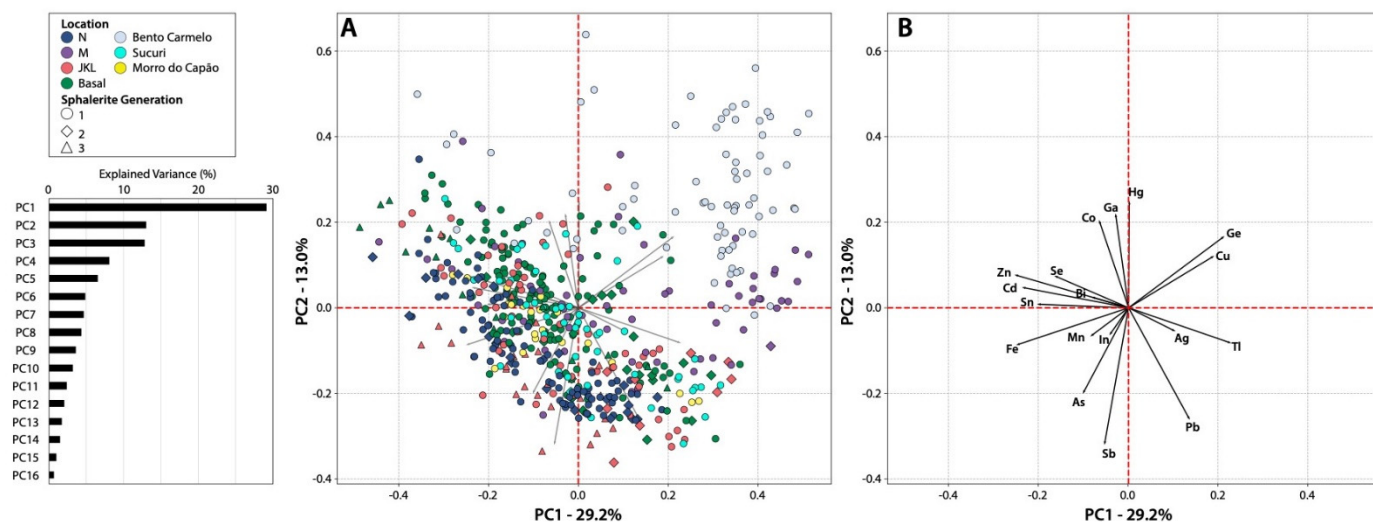


Figure 11. Plot of spherulite analyses in PC1 (29.2%) vs. PC2 (13.0%) (A) and the element components (B) with the eigenvalues of the PCs on the left.

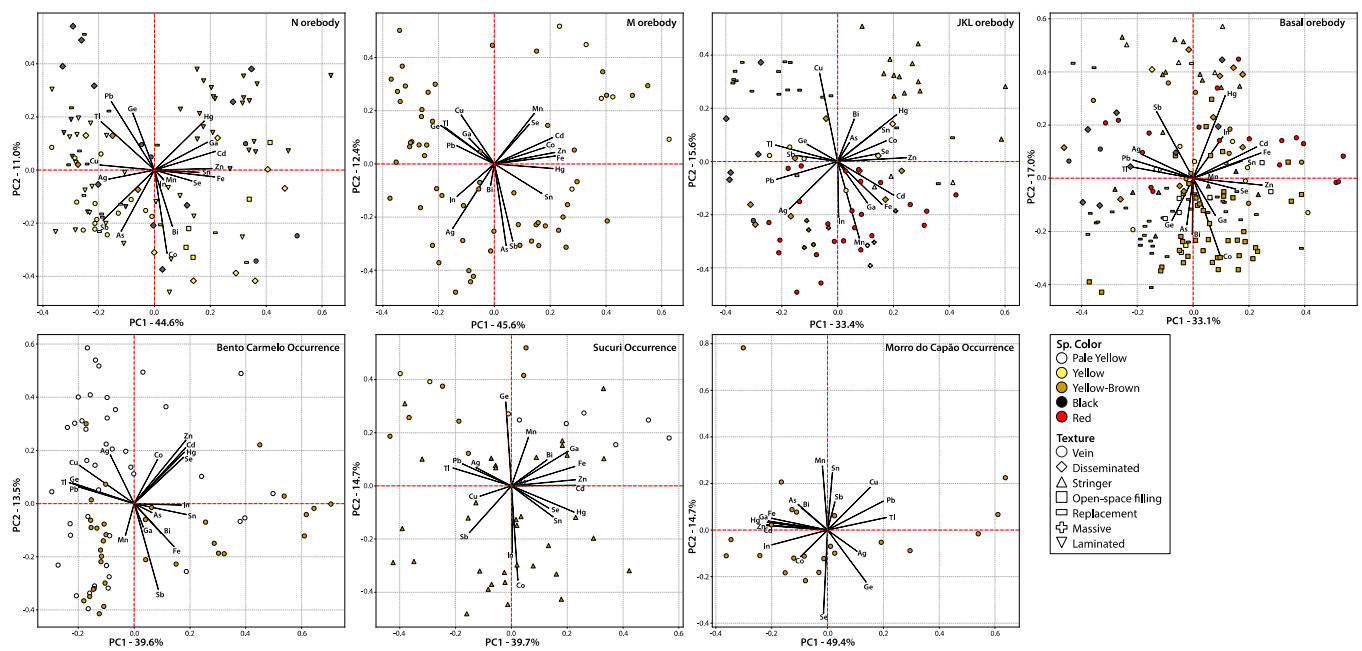


Figure 12. Principal components (PC1 vs. PC2) of sphalerite mineral chemistry from the various orebodies of the Morro Agudo deposit and minor occurrences in the district.

Using a subset of the elemental dataset containing only the elements Ag, Co, Cu, Cd, Fe, Ga, Ge, In, and Mn (which will be applied to geothermometry), PC1 and PC2 capture 52.5% of the variability in the dataset (Figure 13). Sphalerite from JKL, Basal, Sucuri and Morro do Capão plotted mostly in PC1– and PC2–, which are loaded by Ga and In, with the M orebody in PC1+ and PC2– loaded by Ag. The N orebody and Bento Carmelo sphalerite yielded PC2+ values, however, separated by PC1: N orebody is located in PC1– (loaded by Cd, Co, and Mn) and Bento Carmelo is in PC1+ (loaded by Cu and Ge) (Figure 13).

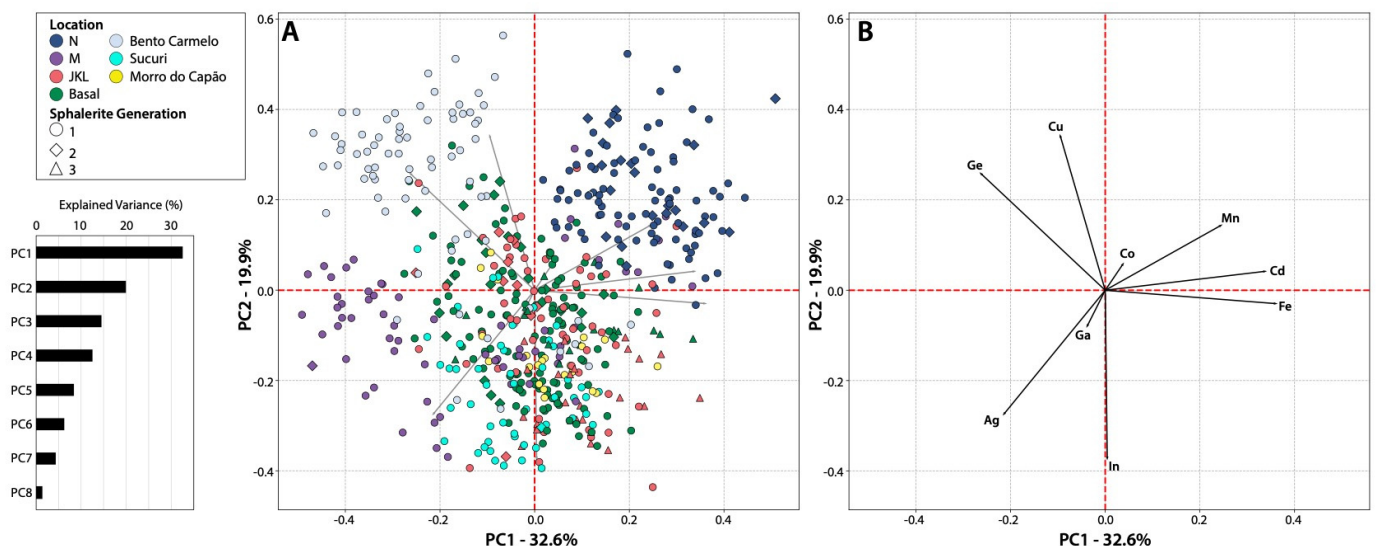


Figure 13. Principal components (PCs) for the sphalerite mineral chemistry using a reduced dataset from the Morro Agudo district (according to Frenzel et al. [27] for geothermometric applications); (A) biplot of the PC1 (32.6%) vs. PC2 (19.9%) plane displaying the spot analyses; (B) elements located in PC1 and PC2.

4.2. Galena Trace Element Content

A total of 170 galena spots were analyzed (Basal: 16; JKL: 69; M: 39; N: 14; Sucuri: 12; Morro do Capão: 20) by LA-ICPMS, and the results are reported in Table 4 (by occurrence) and Table 5 (by generation). The elements commonly detected (>60%) in the analyzed samples were Ag, Bi, Cd, Hg, Sb, Se, Sn and Tl. Arsenic, Co, Cu, Fe, Ga, Ge, In, Mn, Ni, and Zn were also analyzed but were below the detection limit for most of the points as described in Section 3.3. Boxplots for the most commonly detected trace elements are presented by orebodies and occurrences in Figure 14.

Table 4. Trace element contents of galena determined by LA-ICP-MS from the Morro Agudo district. All units are in ppm.

Elements	Basal (n = 16)				JKL (n = 69)				M (n = 39)			
	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD
Ag	109	166	142	100	0.22	104	26.6	100	5.38	14.1	9.16	100
Bi	0.17	3.95	1.94	100	0.04	0.85	0.08	100	0.04	0.71	0.10	100
Cd	35.2	116	71.5	100	27.7	196	61.4	100	22.7	54.3	33.5	100
Cu	-	-	-	-	-	-	-	-	0.44	3.83	1.46	77
Fe	4.76	90.3	23.4	56	-	-	-	-	-	-	-	-
Ge	0.05	0.13	0.09	50	-	-	-	-	-	-	-	-
Hg	0.47	4.03	1.80	100	0.06	0.63	0.23	93	0.09	0.61	0.31	100
Mn	-	-	-	-	0.09	310	25.6	58	0.10	119	10.7	64
Sb	201	292	256	100	0.05	170	65.9	81	44.0	71.7	54.3	100
Se	6.19	31.1	18.1	100	0.77	5.57	1.88	48	0.94	2.92	1.71	49
Sn	0.73	2.4	1.74	100	0.31	1.30	0.68	84	0.38	1.23	0.71	95
Tl	2.26	3.18	2.77	100	0.79	3.82	1.93	100	1.62	2.33	1.91	100

Elements	N (n = 14)				Sucuri (n = 12)				Morro do Capão (n = 20)			
	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD
Ag	0.86	2.04	1.26	100	91.0	114	106	100	32.8	48.8	40.3	100
Bi	0.02	0.06	0.04	100	0.03	0.06	0.05	100	1.75	3.47	2.78	100
Cd	27.0	37.8	31.4	100	15.9	30.9	23.6	100	51.2	85.2	64.1	100
Cu	-	-	-	-	0.02	1.87	0.50	67	0.25	3.64	1.54	65
Fe	-	-	-	-	-	-	-	-	-	-	-	-
Ge	-	-	-	-	-	-	-	-	0.37	0.61	0.50	100
Hg	0.59	4.45	1.77	100	0.12	0.30	0.22	100	0.30	1.31	0.62	100
Mn	0.07	3.75	0.69	86	-	-	-	-	-	-	-	-
Sb	30.6	39.8	34.3	100	113	152	130	100	58.5	88.2	73.1	100
Se	1.92	4.12	1.66	86	1.92	3.80	2.56	100	148	177	159	100
Sn	0.52	1.11	0.75	93	0.42	0.86	0.73	92	0.44	0.92	0.69	95
Tl	3.29	4.82	3.88	100	0.42	0.67	0.54	100	2.74	5.41	3.82	100

Table 5. Trace element contents of galena determined by LA-ICP-MS for the different generations. All units are in ppm.

Elements	1st (n = 141)				2nd (n = 29)			
	Min.	Max.	Mean	% > LoD	Min.	Max.	Mean	% > LoD
Ag	0.22	166	43.1	100	6.69	35.5	16.4	100
Bi	0.02	3.95	0.66	100	0.04	0.85	0.14	100
Cd	15.9	116	52.6	100	25.2	196	44.3	100
Cu	0.28	4.31	1.34	44	0.29	2.49	1.11	69
Fe	-	-	-	-	-	-	-	-
Ge	0.04	0.61	0.37	32	0.05	0.06	0.06	12
Hg	0.07	4.45	0.64	96	0.06	0.61	0.29	100
Mn	0.07	310	14.1	50	0.10	183	20.0	55
Sb	0.05	292	90.3	91	44.7	94.4	63.4	100
Se	0.77	177	37.0	70	0.97	2.92	1.81	52
Sn	0.31	2.40	0.82	90	0.41	1.23	0.76	93
Tl	0.42	5.41	2.31	100	1.80	3.83	2.21	100

The Basal orebody, Sucuri and Morro do Capão occurrences have the highest mean concentrations of Ag (142, 106, and 40.3 ppm, respectively), and Sb (256, 130, and 73.1 ppm); Basal and Morro do Capão yielded the highest Bi (1.94 and 2.78 ppm) and Se mean values (18.1 and 159 ppm); Basal, N orebodies and Morro do Capão have the highest mean Tl (2.77,

3.88, and 3.82 ppm) and Hg (1.80, 1.77, and 0.62 ppm) contents; Basal, JKL orebodies and Morro do Capão the highest Cd (71.5, 61.4, and 64.1 ppm) mean values; and Basal orebody has the highest Sn values (1.74 ppm) (Table 4 and log values plotted in Figure 14).

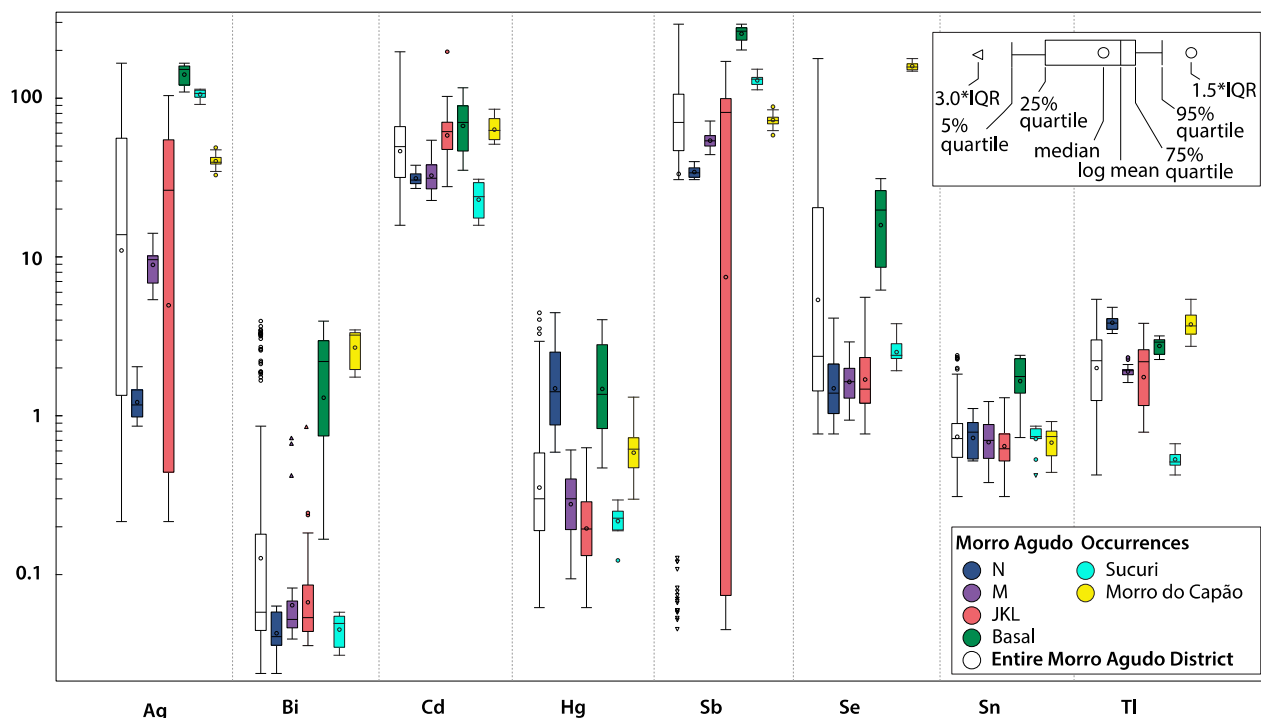


Figure 14. Box-and-whisker plots of common trace elements in galena within the Morro Agudo deposit and district occurrences. All units are in ppm. Note the logarithmic scale and corresponding mean value.

The JKL orebody has the greatest variation in Ag and Sb values, ranging from 0.22 to 104 and 0.05 to 170 ppm, respectively (Figure 14). Morro do Capão also has detectable quantities of Ge in all samples with a mean of 0.50 ppm. Manganese is detectable (>55%) in the N (mean 0.69 ppm), M (10.7 ppm), and JKL (25.6 ppm) orebodies, and Cu is detectable (>55%) in the M (mean 1.46 ppm) orebody and Sucuri (0.50 ppm) and Morro do Capão (1.54 ppm) occurrences (Table 4). The second generation of galena (Gn-II) has lower mean contents of Ag (Gn-I: 43.1 ppm; Gn-II: 16.4 ppm), Cd (52.6 and 44.3 ppm), and Se (37.0 and 1.81 ppm), but higher mean concentration of Mn (14.1 and 20.0 ppm) than Gn-I (Table 5).

PCA of Galena

Trace elements (Ag, Bi, Cd, Hg, Sb, Se, Sn, and Tl) that were detectable in >60% of analyses were used for PCA (Figure 15). The first two principal components account for ~75% (PC1 = 49.8%; PC2 = 25.8%) of the elemental variability, separating into four clusters: (1) Morro do Capão in PC1–, PC2– with high loadings of Se; (2) samples from the N orebody with high loadings of Cd, Hg, Sn and Tl located in PC1+, PC2+; (3) JKL samples situated in PC1+, PC2– with loadings of Bi; and (4) encompassing all of the Sucuri occurrence, M orebody, and a second JKL population located in PC1– and PC2+, loaded by Ag and Sb (Figure 15).

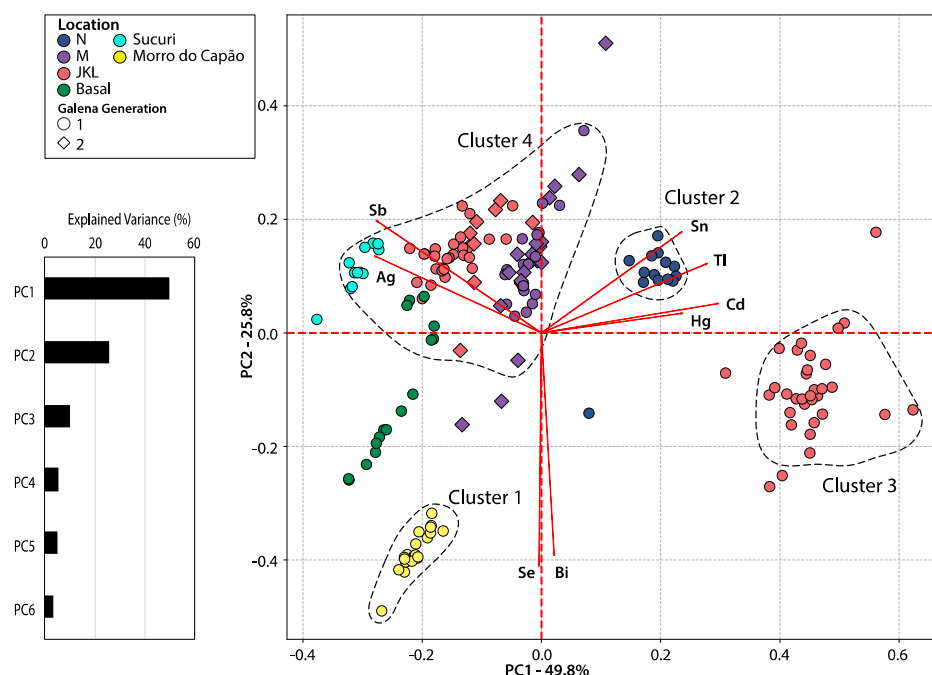


Figure 15. Biplot of PC1 (49.8%) and PC2 (25.8%) of the galena mineral chemistry from the Morro Agudo district with associated eigenvalues (explained variance).

5. Discussion

5.1. Ore-Forming Temperatures

Due to the lack of fluid inclusions suitable for microthermometric investigation in the collected samples, the temperatures of sphalerite formation were calculated following the methodology of Frenzel et al. [27], which utilizes the concentrations of Ga, Ge, In, Mn, and Fe in sphalerite to estimate the temperature, and the results are presented in Figure 16 and Table 6. Sphalerite yielded similar, typically hot temperature ranges, with the mean slightly higher in the N orebody (range: 186–320 °C; mean: 247 °C), followed by Morro do Capão (222–294 °C; 240 °C), Sucuri (167–294 °C; 238 °C), JKL (119–297 °C; 234 °C) and Basal orebodies (147–297 °C; 227 °C). The M orebody (113–294 °C; 193 °C) and Bento Carmelo (82–262 °C; 168 °C) occurrence yielded the lowest mean formation temperatures (Figure 16A; Table 6).

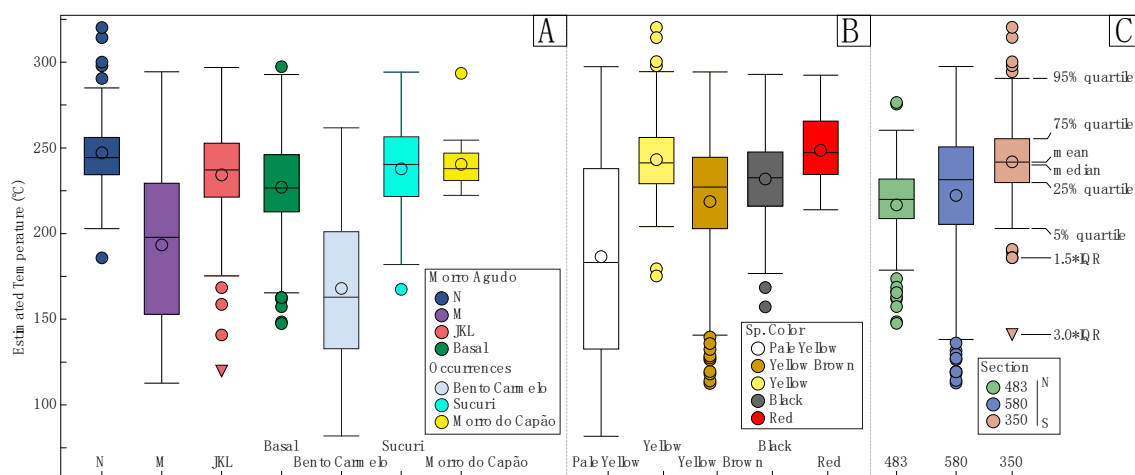


Figure 16. Estimated formation temperatures of sphalerite after Frenzel et al. [27] distinguished by orebody (A), sphalerite color (B), and section in the Morro Agudo deposit (C).

Table 6. Estimated temperatures of the Morro Agudo deposit and Sucuri, Bento Carmelo, and Morro do Capão occurrences compared with fluid inclusion data from the Morro Agudo deposit [4]. (n) = number of samples. Blocks are defined by their proximity to the principal fault, with block A being the most proximal (see Figure 4).

Orebody/Occurrence (n)	Estimate Temperature (°C)				
	Min.	Max.	Mean	Mode	Std. Dev.
N (108)	186	320	247	240–246	21
M (63)	113	294	193	213–222	46
JKL (88)	119	297	234	226–235	31
Basal (165)	147	297	227	222–230	28
Bento Carmelo (74)	82	262	168	199–208	43
Sucuri (50)	167	294	238	237–244	25
Morro do Capão (22)	222	294	240	237–240	15

Sphalerite Color (n)	Estimate Temperature (°C)				
	Min.	Max.	Mean	Mode	Std. Dev.
Pale Yellow (67)	82	297	187	233–243	58
Yellow (85)	175	320	243	241–248	26
Yellow-Brown (323)	113	294	219	231–240	37
Black (52)	157	293	232	225–239	28
Red (43)	214	292	249	230–234	18

Orebody	Estimate Temperature (°C)—Cunha et al. (2000)			
	Block (n)	Min.	Max.	Mode.
N	C (16)	120	150	135
M	A (74)	100	164	135
	A (153)	100	300	165
JKL	B (15)	140	160	155
	C (65)	80	170	155
Basal	B (24)	80	210	165

These calculated temperatures are in accordance with the reported range of fluid inclusion homogenization temperatures (80–300 °C: [4]) for the Morro Agudo deposit and with temperatures determined from the fractionation of sulfur in cogenetic sphalerite-galena pairs, which ranged from 105 to 256 °C [16]. These estimated temperatures of formation based on trace element chemistry suggest lower pressures and shallower depths during ore formation and are supported by the ore textures [15–17]. However, fluid inclusion results show the highest temperatures occur in JKL (80–300 °C, mode: 155–165 °C) and the lowest in the N orebody (100–150 °C, mode 135 °C: [4]). No evidence of proximal-distal geothermal gradient or zonation in the orebodies in the studied sections (proximal: 350 and more distal: 483 and 580) could be determined to support the hypothesis by Cunha et al. [4] and Misi et al. [16] that the principal fault acted as the main fluid conduit during mineralization (Figures 4 and 16A; Table 6). However, calculated trace-element temperatures generally increased southward in the Morro Agudo sections (from 483, 580 to 350; Figure 16C).

5.2. Implications to Ore Genesis in the Morro Agudo District

Aldis et al. [15] proposed that the formation of the Morro Agudo district, especially within the Morro Agudo deposit and Sucuri occurrence, is the result of progressive fluid mixing of saline, Fe- and Cd-bearing fluids with little influence from the host rock compositions. In the Morro Agudo deposit and Sucuri, the data show overlaps in trace element composition among various sphalerite colors and in multiple locations (Figures 8 and 9; Tables 2 and 3), and the PCA (Figures 11 and 12) assists in further constraining the complex composition of the various mineralizing fluids. The various generations of sphalerite from several locations have Fe loadings associated with As-In-Mn-Sb (PC2– and PC1–), and

with Ag-Pb-Tl (PC2– and PC1+), whereas Cd loadings are associated with Bi-Co-Ga-Se-Sn (PC2+, PC1–) (Figure 11). Moreover, the Cu-Ge-Hg association (in PC1+ and PC2+) relates mainly to sphalerite from the Bento Carmelo occurrence and a few samples from the Basal and M orebodies. These elemental data associations and the wide range of sphalerite formation temperatures imply that fluids from distinct sources and temperatures interacted in zones that formed economic concentrations of sphalerite. Significantly, most of the Bento Carmelo sphalerite and some from M orebodies yielded the lowest mean temperatures for sphalerite formation (168–193 °C) compared to the other orebodies and occurrences (227–240 °C), suggesting that the Cu-Ge-Hg-bearing fluids were cooler (Figure 16A; Table 6).

Therefore, based on the sphalerite trace element data, assuming the trace element mineral chemistry of sphalerite equals the relative abundances in ore-forming fluids, it is tempting to propose that Zn and Pb mineralizing fluids derive from at least two major distinct sources: one as the major fluid/rock source for Fe, As, In, Mn, Sb, Ag, Tl; and another as the main source for Cd, Bi, Co, Ga and Se. The enrichments of Ge, Hg and Cu primarily associated with Bento Carmelo could be related to a distinct localized fluid/metal source. Interestingly, for the elements that were detected in galena, Ag and Sb are associated with PC1– (comprising most of the analyzed points), and Cd and Bi are associated with PC1+, which supports the interpretation of at least two participating mineralizing fluids.

5.3. Comparison with Vazante-Paracatu Mineral Belt

The composition of sphalerite in the Morro Agudo district reported in this study has higher concentrations of Fe (mean 1.28 wt.%), similar Cd (3759 ppm) contents, but significantly lower Ag (4.95 ppm), Cu (40 ppm), and Ge (7.19 ppm) compared with Fagundes (Fe: 0.46 wt.%, Ag: 210 ppm, Cd: 3350 ppm, Cu: 300 ppm, Ge: 540 ppm), Ambrósia (Fe: 0.78 wt.%, Ag: 120 ppm, Cd: 1190 ppm, Ge: 300 ppm), and Vazante (Fe: 0.09 wt.%, Ag: 120 ppm, Cd: 8410 ppm, Cu: 240 ppm, Ge: 130 ppm) (Figure 17; Tables 2 and 7). Red sphalerite (interpreted to be the last stage) in the Morro Agudo district yielded elevated contents of Fe (Figure 9; Table 3), which is also evident from late-stage sphalerite in Ambrósia and Fagundes (Figure 17; Table 6) [6]. However, late sphalerite at the Ambrósia deposit is also enriched in Cd.

Table 7. Previously reported mean trace element compositions (Fe in wt.% and other elements in ppm) of sphalerite and galena from the Fagundes, Ambrósia, and Vazante deposits. Data from Monteiro et al. [6] and references therein.

Mineral	Deposit	Ag	Cd	Co	Cu	Fe	Ga	Ge	Mn	Zn
Galena	Ambrósia	140	420	-	-	-	3340	7010	-	1610
	Fagundes	170	420	-	-	-	3470	4970	-	740
	Vazante	230	710	-	-	-	120	380	-	3920
	Morro Agudo	71	-	-	41	-	10	-	-	-
Sphalerite	Ambrósia	120	1190	-	-	0.78	-	300	-	-
	Fagundes	210	3350	-	300	0.46	-	540	-	-
	Vazante	120	8410	-	240	0.09	-	130	-	-
	Morro Agudo	20	7700	10	72	0.64	10	-	825	-

Galena in the Morro Agudo district has significantly lower concentrations of Ag (mean 39 ppm), Cd (51 ppm), Cu (1.11 ppm), Ga (0.11 ppm), Ge (0.27 ppm), and Zn (681 ppm) compared to elevated contents observed in Fagundes (Ag: 170 ppm, Cd: 420 ppm, Ga: 3470 ppm, Ge: 4970 ppm, Zn: 740 ppm), Ambrósia (Ag: 140 ppm, Cd: 420 ppm, Ga: 3340 ppm, Ge: 7010 ppm, Zn: 1610 ppm), and Vazante (Ag: 230 ppm, Cd: 710 ppm, Ga: 120 ppm, Ge: 380 ppm, Zn: 3920 ppm) deposits (Table 7). The lower concentrations in various trace elements (i.e., Ag, Co, Cu, Ga, Ge, In) throughout the Morro Agudo district and relatively higher contents in other Vazante carbonate-hosted Pb-Zn sulfide deposits (Fagundes, Ambrósia, Vazante: Table 7) within the belt suggest distinct rock/fluid sources for this suite of elements.

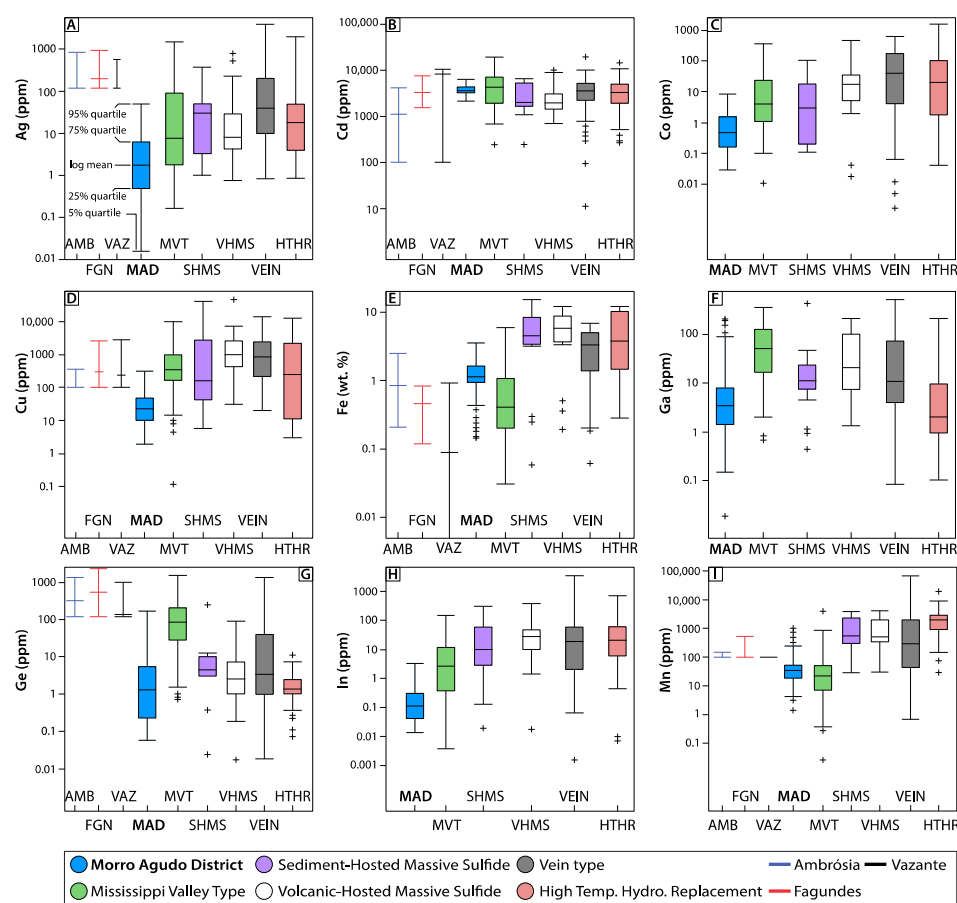


Figure 17. Box-and-whisker plots of minor and trace elements in sphalerite (A) silver; (B) cadmium; (C) cobalt; (D) copper; (E) iron; (F) gallium; (G) germanium; (H) indium; and (I) manganese within the Morro Agudo district compared with known compositional ranges of elements for Mississippi Valley-type (MVT), Sediment-Hosted Massive Sulfide (SHMS), Volcanic-Hosted Massive Sulfide (VHMS), Vein-type (VEIN), or High-Temperature Hydrothermal Replacement (HTHR) deposits (Frenzel et al. [27] and references therein) and other zinc deposits in the Vazante-Paracatu Mineral Belt [6]. Note the logarithmic scale and corresponding mean value.

Various authors (e.g., [1,6,7,16]) have proposed that multiple pulses of metalliferous fluids formed the Zn-Pb sulfide and zinc silicate deposits in the Vazante basin during the Brasiliano orogeny. However, the timing of mineralization throughout the basin is uncertain, and further work is required to constrain the chronology. The controversial Pb and S isotope compositions of galena from the Morro Agudo district [5,16,18,19] could be explained by the multiple sources of fluids involved in the epigenetic mineralization associated with the Brasiliano orogenic event rather than distinct events (syn-depositional and syn-tectonic). Further investigation integrating mineral paragenesis, major and trace elemental compositions, and isotopic studies would be required to further constrain the age of mineralization.

Botura Neto and Filho [48] noted the presence of both extensional and compressive stress regimes in the Vazante Group based on high-angle veins and normal faulting and low-angle veins and reverse faulting, respectively. The association of these extensional and contractional structures with hydrothermal minerals and mineralization during the Brasiliano orogeny suggests multiple phases of mineralization [48]. The similarities in ore textures (e.g., open-space filling, veins, breccias) and strong structural control (i.e., high angle faults and structures) indicates ore deposition was similar at Morro Agudo. The presence of multiple generations of sulfide, carbonate, and silicate minerals [6,7,15,17,48] and progressive increase in temperature between sphalerite generations in the Morro Agudo

district (Figure 16, Table 6) is also evident in the Fagundes and Ambrósia deposits [6,59], indicating episodic migration of warmer fluids in the late episodes of mineralization during the Brasiliano orogeny.

5.4. Deposit Model

The elemental compositions of sphalerite from different deposit types (volcanogenic and sediment-hosted massive sulfide, vein-type, high-temperature hydrothermal replacement, and MVT: [27]) are compared with the results of the Morro Agudo district (Figure 17). Gallium and Ge are expected to be enriched in low-temperature systems, whereas high-temperature magmatic-hydrothermal systems, VMS, and SEDEX deposits typically have high Fe (>5–10%), In (>100 ppm) and Mn (>1000 ppm) contents [21,22,27,30,32,36,38]. Silver, Cd, Co, and Cu are typically temperature-independent, whereas As, Hg, Sb, and Tl form in low-temperature systems [27,34]. The low concentrations of Fe (1.28%), In (0.24 ppm), and Mn (62.6 ppm) within the Morro Agudo district correspond well with known ranges for MVT deposits, but the Ga (8.30 ppm) and Ge (9.46 ppm) contents are lower than expected in low-temperature systems (Figure 17).

The trace element composition of sphalerite from the Morro Agudo district more closely resembles those of sphalerite in MVT deposits than of sphalerite in Irish-type or SEDEX deposits. However, the calculated sphalerite formation temperatures align well with known fluid inclusions of Irish-type (70–280 °C) or SEDEX (70–300 °C) models rather than lower temperature MVT (90–150 °C) systems [60–62]. These elevated fluid temperatures may reflect a higher geothermal gradient or vigorous advective heat transport related to the uplift during the Brasiliano orogeny and topographic-driven fluid flow [61,62] as the mineralogical, textural, and fluid composition evidence support mineralization analogous to MVT. This contrasts with previous interpretations that have classified the Morro Agudo district as SEDEX or Irish-type [2,3,5,50].

6. Conclusions

1. The earliest generations of pale yellow to yellow-brown sphalerite (interpreted as the first generation: Sp-I) have elevated contents of Co, Cu, Ge, Hg, Pb, Tl and lower concentrations of Fe and Mn compared to the red generation of sphalerite (interpreted as the last generation: Sp-III), and the first generation of galena (Gn-I) is enriched in Ag, Cd, and Se and depleted in Cu and Mn relative to the second generation of galena (Gn-II).
2. The PCA analysis of sphalerite and galena composition suggests that the mineralizing fluids were associated with two compositions/sources enriched in: (1) Fe, As, In, Mn, Sb, Ag, Tl; and (2) Cd, Bi, Co, Ga, and Se.
3. The broad overlap in concentrations between sphalerite generations and locations in the Morro Agudo district suggests the mixing of fluids from these distinct sources. Bento Carmelo is enriched in Cu, Ge, and Hg, which supports previous interpretations that mineralizing processes are derived from unique fluid/sources.
4. Sphalerite and galena in the Morro Agudo district have lower concentrations of the trace elements Ag, Cu, Ge and Ag, Cd, Cu, Ga, Ge, and Zn, respectively, compared to sphalerite and galena from other deposits in the Vazante Group.
5. The characteristically low Fe, In, and Mn contents in sphalerite from the Morro Agudo district and other Vazante-hosted Zn-Pb sulfide deposits are more similar to MVT deposits than SEDEX or Irish-type deposits.
6. Sphalerite geothermometry suggests the Morro Agudo district mineralization formed from fluids with highly variable temperatures ranging from 82 to 320 °C (mean temperatures ~200–250 °C).
7. The higher formation temperatures of the Morro Agudo deposits compared to typical MVT deposits may reflect a high geothermal gradient or high advective heat transport caused by the Brasiliano orogeny.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/min12081028/s1>. Table S1. Dwell times for elements analyzed by LA-ICP-MS for both galena and sphalerite. Table S2. Sphalerite and galena LA-ICP-MS data; Figure S1. Plot of sphalerite analyses in PC1 (29.2%) vs. PC3 (12.8%) and associated element components. Figure S2. Plot of sphalerite analyses in PC1 (29.2%) vs. PC4 (8.1%) and associated element components. Figure S3. Plot of sphalerite analyses in PC2 (13.0%) vs. PC3 (12.8%) and associated element components. Figure S4. Plot of sphalerite analyses in PC2 (13.0%) vs. PC4 (8.1%) and associated element components. Figure S5. PC1 vs. PC2 of sphalerite analyses showing the distribution from the Morro Agudo District by (A) section of the Morro Agudo deposit, and (B) host rock type.

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